LINEAR ALGEBRA

Underwood Dudley

Contents

1 Vector Spaces **5**
- 1.1 Definitions 5
- 1.2 Bases 6
- 1.3 Dimension of a Vector Space 7
- 1.4 Sums and Direct Sums 8

2 Matrices **9**
- 2.1 The Space of Matrices 9
- 2.2 Linear Equations 10
- 2.3 Multiplication of Matrices 10

3 Linear Mappings **13**
- 3.1 Mappings 13
- 3.2 Linear Mappings 14
- 3.3 The Kernel and Image of a Linear Map 14
- 3.4 Composition and Inverse of Linear Mappings 15
- 3.5 Geometric Applications 16

4 Linear Maps and Matrices **17**
- 4.1 The Linear Map Associated with a Matrix 17
- 4.2 The Matrix Associated with a Linear Map 17
- 4.3 Bases, Matrices, and Linear Maps 17

5 Scalar Products and Orthogonality **19**
- 5.1 Scalar Products 19
- 5.2 Orthogonal Bases, Positive Definite Case 20
- 5.3 Application to Linear Equations; The Rank 21
- 5.4 Bilinear Maps and Matrices 21
- 5.5 General Orthogonal Bases 21
- 5.6 The Dual Space and Scalar Products 22
- 5.7 Quadratic Forms 22
- 5.8 Sylvester's Theorem 22

6 Determinants **25**
- 6.1 Determinants of Order 2 25

6.2 Existence of Determinants . . . 26
6.3 Additional Properties of Determinants . . . 26
6.4 Cramer's Rule . . . 27
6.5 Permutations . . . 27
6.6 Expansion Formula and Uniqueness of Determinants . . . 28
6.7 Inverse of a Matrix . . . 28
6.8 The Rank of a Matrix and Subdeterminants . . . 28

7 Symmetric, Hermitian, and Unitary Operators . . . 29
7.1 Symmetric Operators . . . 29
7.2 Hermitian Operators . . . 29
7.3 Unitary Operators . . . 30

8 Eigenvectors and Eigenvalues . . . 33
8.1 The Characteristic Polynomial . . . 33
8.2 Eigenvectors and Eigenvalues . . . 33
8.3 Eigenvalues and Eigenvectors of Symmetric Matrices . . . 33
8.4 Diagonalization of a Symmetric Linear Map . . . 34
8.5 The Hermitian Case . . . 34
8.6 Unitary Operators . . . 35

9 Polynomials and Matrices . . . 37
9.1 Polynomials . . . 37
9.2 Polynomials of Matrices and Linear Maps . . . 37

10 Triangulation of Matrices and Linear Maps . . . 39
10.1 Existence of Triangulation . . . 39
10.2 Theorem of Hamilton-Cayley . . . 40
10.3 Diagonalization of Unitary Maps . . . 40

11 Polynomials and Primary Decomposition . . . 43
11.1 The Euclidean Algorithm . . . 43
11.2 Greatest Common Divisor . . . 43
11.3 Unique Factorization . . . 44
11.4 Application to the Decomposition of a Vector Space . . . 44
11.5 Schur's Lemma . . . 45
11.6 The Jordan Normal Form . . . 45

12 Convex Sets . . . 47
12.1 Definitions . . . 47
12.2 Separating Hyperplanes . . . 47
12.3 Extreme Points and Supporting Hyperplanes . . . 48
12.4 The Krein-Milman Theorem . . . 48

A Complex Numbers **49**

B Iwasawa Decomposition and Others **51**

B.1 Iwasawa Decomposition 51

B.2 Polar Decompositions 52

1

VECTOR SPACES

1.1 Definitions

DEFINITION 1.1. Field A field K is a set with two operations (addition and multiplication) satisfying the following axioms for all elements $x, y, z \in K$:

(1) $(x + y) + z = x + (y + z)$ and $(xy)z = x(yz)$ (associativity).

(2) $x + y = y + x$ and $xy = yx$ (commutativity).

(3) $x(y + z) = xy + xz$ (distributivity).

(4) There exist elements $0, 1 \in K$ with $0 \neq 1$ such that $x + 0 = x$ and $1 \cdot x = x$ for all x.

(5) For each x there exists an element $-x$ such that $x + (-x) = 0$.

(6) If $x \neq 0$, there exists an element x^{-1} such that $xx^{-1} = 1$.

DEFINITION 1.2. Vector Space Let K be a field. A vector space over K is a set V with two operations (addition of vectors and scalar multiplication) satisfying the following axioms (where $u, v, w \in V$ and $a, b \in K$):

(VS 1) $(u + v) + w = u + (v + w)$.

(VS 2) $u + v = v + u$.

(VS 3) There exists an element $O \in V$ such that $u + O = u$.

(VS 4) For each u there exists $-u$ such that $u + (-u) = O$.

(VS 5) $a(u + v) = au + av$.

(VS 6) $(a + b)u = au + bu$.

(VS 7) $(ab)u = a(bu)$.

(VS 8) $1u = u$.

Elements of V are called vectors, and elements of K are called scalars.

THEOREM 1.1. Uniqueness of Zero Vector In any vector space, the zero vector is unique.

Proof. Suppose O and O' are both zero vectors. Then:

$$O = O + O' = O' + O = O'$$

Thus $O = O'$, proving uniqueness. □

THEOREM 1.2. Properties of Scalar Multiplication In any vector space over K:

(a) $0u = O$ for all $u \in V$.

(b) $aO = O$ for all $a \in K$.

(c) $(-1)u = -u$ for all $u \in V$.

Proof. (a) We have $0u = (0+0)u = 0u + 0u$. Adding $-0u$ to both sides gives $O = 0u$.
(b) We have $aO = a(O + O) = aO + aO$. Adding $-aO$ to both sides gives $O = aO$.
(c) We have $u + (-1)u = 1u + (-1)u = (1 + (-1))u = 0u = O$. Thus $(-1)u = -u$. □

DEFINITION 1.3. Subspace Let V be a vector space over K. A subset W of V is called a subspace if:

(1) $O \in W$.

(2) If $u, v \in W$, then $u + v \in W$.

(3) If $u \in W$ and $a \in K$, then $au \in W$.

DEFINITION 1.4. Linear Combination Let V be a vector space over K and let $v_1, \ldots, v_n \in V$. An expression of the form

$$a_1v_1 + a_2v_2 + \cdots + a_nv_n$$

with $a_i \in K$ is called a linear combination of $v_1, \ldots, v_n$.

DEFINITION 1.5. Span Let $v_1, \ldots, v_n \in V$. The set of all linear combinations of $v_1, \ldots, v_n$ is called the span of $\{v_1, \ldots, v_n\}$, denoted by $\text{span}(v_1, \ldots, v_n)$.

THEOREM 1.3. Span is a Subspace The span of any finite set of vectors in V is a subspace of V.

Proof. Let $W = \text{span}(v_1, \ldots, v_n)$.
(1) $O = 0v_1 + \cdots + 0v_n \in W$.
(2) If $u = a_1v_1 + \cdots + a_nv_n$ and $w = b_1v_1 + \cdots + b_nv_n$ are in W, then:

$$u + w = (a_1 + b_1)v_1 + \cdots + (a_n + b_n)v_n \in W$$

(3) If $u = a_1v_1 + \cdots + a_nv_n \in W$ and $c \in K$, then:

$$cu = (ca_1)v_1 + \cdots + (ca_n)v_n \in W$$

Thus W is a subspace. □

1.2 Bases

DEFINITION 1.6. Linearly Independent Vectors $v_1, \ldots, v_n$ in a vector space V are said to be linearly independent if whenever

$$a_1v_1 + \cdots + a_nv_n = O$$

with $a_i \in K$, then $a_1 = a_2 = \cdots = a_n = 0$. Otherwise, they are linearly dependent.

DEFINITION 1.7. Basis A set $\{v_1, \ldots, v_n\}$ of vectors in V is called a basis of V if:

(1) $v_1, \ldots, v_n$ are linearly independent.

(2) Every element of V can be expressed as a linear combination of $v_1, \ldots, v_n$.

THEOREM 1.4. Uniqueness of Representation If $\{v_1, \ldots, v_n\}$ is a basis of V, then every element $v \in V$ can be written uniquely as

$$v = a_1 v_1 + \cdots + a_n v_n$$

with $a_i \in K$.

Proof. Since $\{v_1, \ldots, v_n\}$ is a basis, every $v \in V$ can be written as such a linear combination. Suppose v has two representations:

$$v = a_1 v_1 + \cdots + a_n v_n = b_1 v_1 + \cdots + b_n v_n$$

Then:

$$(a_1 - b_1)v_1 + \cdots + (a_n - b_n)v_n = O$$

By linear independence, $a_i - b_i = 0$ for all i, so $a_i = b_i$. □

DEFINITION 1.8. Coordinate Vector Let $\{v_1, \ldots, v_n\}$ be a basis of V and let $v \in V$ with $v = a_1 v_1 + \cdots + a_n v_n$. The scalars $a_1, \ldots, a_n$ are called the coordinates of v with respect to this basis, and the column vector $(a_1, \ldots, a_n)^T$ is called the coordinate vector of v.

THEOREM 1.5. Extension to a Basis Let V be a vector space and let $v_1, \ldots, v_r$ be linearly independent elements of V. If V is generated by $v_1, \ldots, v_r, w_1, \ldots, w_s$, then we can select a subset of $\{w_1, \ldots, w_s\}$ which together with $v_1, \ldots, v_r$ forms a basis of V.

Proof. We prove this by induction. If w_1 is a linear combination of $v_1, \ldots, v_r$, then V is generated by $v_1, \ldots, v_r, w_2, \ldots, w_s$. If w_1 is not a linear combination of $v_1, \ldots, v_r$, then $\{v_1, \ldots, v_r, w_1\}$ is linearly independent. We continue this process, adding w_i to our set if it is not a linear combination of the vectors already selected. After at most s steps, we obtain a basis. □

1.3 Dimension of a Vector Space

THEOREM 1.6. Basis Size Invariance Let V be a vector space having a basis consisting of n elements. Then any set of linearly independent elements of V has at most n elements.

Proof. Let $\{v_1, \ldots, v_n\}$ be a basis of V. Suppose $\{w_1, \ldots, w_m\}$ is linearly independent with $m > n$. Since V is generated by $\{v_1, \ldots, v_n\}$, by the extension theorem we can extend $\{w_1, \ldots, w_m\}$ to a basis. But any basis has exactly n elements, contradicting $m > n$. □

DEFINITION 1.9. Dimension Let V be a vector space having a basis consisting of n elements. We call n the dimension of V, denoted $\dim V$. If V consists of O alone, we define $\dim V = 0$.

THEOREM 1.7. All Bases Have Same Size Any two bases of a finite-dimensional vector space have the same number of elements.

Proof. Let $\{v_1, \ldots, v_n\}$ and $\{w_1, \ldots, w_m\}$ be bases of V. Since $\{v_1, \ldots, v_n\}$ is a basis and $\{w_1, \ldots, w_m\}$ is linearly independent, we have $m \leq n$. Similarly, $n \leq m$. Thus $m = n$. □

THEOREM 1.8. Characterization of Basis Let V be a vector space of dimension n. A set of n linearly independent vectors in V is a basis.

Proof. Let $\{v_1, \ldots, v_n\}$ be linearly independent. If this set does not span V, then there exists $v \in V$ not in $\operatorname{span}(v_1, \ldots, v_n)$. Then $\{v_1, \ldots, v_n, v\}$ is linearly independent, contradicting that the maximum size of a linearly independent set is n. □

1.4 Sums and Direct Sums

DEFINITION 1.10. Sum of Subspaces Let U, W be subspaces of a vector space V. The sum $U + W$ is the set of all elements $u + w$ with $u \in U$ and $w \in W$.

THEOREM 1.9. Sum is a Subspace If U, W are subspaces of V, then $U + W$ is a subspace of V.

Proof. (1) $O = O + O \in U + W$.

(2) If $u_1+w_1, u_2+w_2 \in U+W$, then $(u_1+w_1)+(u_2+w_2) = (u_1+u_2)+(w_1+w_2) \in U+W$.

(3) If $u + w \in U + W$ and $a \in K$, then $a(u + w) = au + aw \in U + W$. □

DEFINITION 1.11. Direct Sum Let U, W be subspaces of V. We say that V is the direct sum of U and W, written $V = U \oplus W$, if every element $v \in V$ can be written uniquely as $v = u + w$ with $u \in U$ and $w \in W$.

THEOREM 1.10. Criterion for Direct Sum Let U, W be subspaces of V. Then $V = U \oplus W$ if and only if $V = U + W$ and $U \cap W = \{O\}$.

Proof. ($\Rightarrow$) Suppose $V = U \oplus W$. Clearly $V = U + W$. If $v \in U \cap W$, then $v = v + O = O + v$, so by uniqueness, $v = O$.

($\Leftarrow$) Suppose $V = U+W$ and $U \cap W = \{O\}$. Every $v \in V$ can be written as $v = u+w$. If $v = u_1 + w_1 = u_2 + w_2$, then $u_1 - u_2 = w_2 - w_1 \in U \cap W = \{O\}$. Thus $u_1 = u_2$ and $w_1 = w_2$, proving uniqueness. □

DEFINITION 1.12. Direct Product Let U, W be vector spaces over K. The direct product $U \times W$ is the set of all pairs (u, w) with $u \in U$ and $w \in W$, with componentwise addition and scalar multiplication.

THEOREM 1.11. Dimension of Direct Product If U, W are finite-dimensional vector spaces, then:

$$\dim(U \times W) = \dim U + \dim W$$

Proof. Let $\{u_1, \ldots, u_m\}$ be a basis of U and $\{w_1, \ldots, w_n\}$ be a basis of W. Then the set $\{(u_1, O), \ldots, (u_m, O), (O, w_1), \ldots, (O, w_n)\}$ is a basis of $U \times W$. It has $m+n$ elements, so $\dim(U \times W) = m + n = \dim U + \dim W$. □

2

MATRICES

2.1 The Space of Matrices

DEFINITION 2.1. Matrix Let K be a field. An $m \times n$ matrix in K is a rectangular array of elements of K with m rows and n columns:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

We write $A = (a_{ij})$ where $i = 1, \ldots, m$ and $j = 1, \ldots, n$. The element a_{ij} is called the (i, j)-entry of A.

DEFINITION 2.2. Row and Column Vectors The i-th row of a matrix $A = (a_{ij})$ is the $1 \times n$ matrix $(a_{i1}, a_{i2}, \ldots, a_{in})$. The j-th column is the $m \times 1$ matrix:

$$\begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix}$$

DEFINITION 2.3. Square Matrix A matrix is called square if $m = n$. A square matrix of size $n \times n$ is said to have order n.

DEFINITION 2.4. Matrix Addition Let $A = (a_{ij})$ and $B = (b_{ij})$ be $m \times n$ matrices. Their sum $A + B$ is the $m \times n$ matrix with (i, j)-entry $a_{ij} + b_{ij}$.

DEFINITION 2.5. Scalar Multiplication of Matrices Let $A = (a_{ij})$ be an $m \times n$ matrix and $c \in K$. The product cA is the $m \times n$ matrix with (i, j)-entry ca_{ij}.

THEOREM 2.1. Matrix Space is a Vector Space The set of all $m \times n$ matrices with entries in K forms a vector space over K under matrix addition and scalar multiplication.

Proof. The vector space axioms VS 1–VS 8 follow directly from the corresponding field axioms for K, verified entrywise. □

DEFINITION 2.6. Transpose Let $A = (a_{ij})$ be an $m \times n$ matrix. The transpose of A, denoted ${}^tA$ (or A^T), is the $n \times m$ matrix whose (j, i)-entry is a_{ij}.

DEFINITION 2.7. Symmetric Matrix A square matrix A is symmetric if ${}^tA = A$.

DEFINITION 2.8. Diagonal Matrix A square matrix $A = (a_{ij})$ is diagonal if $a_{ij} = 0$ for all $i \neq j$.

DEFINITION 2.9. Identity Matrix The $n \times n$ identity matrix I_n (or simply I) is

the diagonal matrix with all diagonal entries equal to 1:

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

2.2 Linear Equations

DEFINITION 2.10. System of Linear Equations A system of m linear equations in n unknowns $x_1, \ldots, x_n$ over a field K is:

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

where $a_{ij}, b_i \in K$. The matrix $A = (a_{ij})$ is called the coefficient matrix.

DEFINITION 2.11. Homogeneous System A system of linear equations is homogeneous if $b_1 = b_2 = \cdots = b_m = 0$.

DEFINITION 2.12. Trivial Solution The trivial solution of a homogeneous system is $x_1 = x_2 = \cdots = x_n = 0$.

THEOREM 2.2. Existence of Nontrivial Solutions Let A be an $m \times n$ matrix over K. If $n > m$, then the homogeneous system $AX = O$ has a nontrivial solution.

Proof. The column vectors $A^1, \ldots, A^n$ of A are elements of K^m, which has dimension m. Since $n > m$, these n vectors are linearly dependent. Thus there exist $x_1, \ldots, x_n$ not all zero such that $x_1A^1 + \cdots + x_nA^n = O$, giving a nontrivial solution. □

THEOREM 2.3. Uniqueness of Solution Let $AX = B$ be a system of linear equations. Assume that the column vectors $A^1, \ldots, A^n$ of A are linearly independent. Then the system has at most one solution.

Proof. Suppose X and Y are both solutions. Then $AX = B = AY$, so $A(X-Y) = O$. Writing $X - Y = (z_1, \ldots, z_n)^T$, we have $z_1A^1 + \cdots + z_nA^n = O$. By linear independence, $z_1 = \cdots = z_n = 0$, so $X = Y$. □

2.3 Multiplication of Matrices

DEFINITION 2.13. Matrix Multiplication Let $A = (a_{ij})$ be an $m \times n$ matrix and $B = (b_{jk})$ be an $n \times p$ matrix. Their product AB is the $m \times p$ matrix with (i, k)-entry:

$$c_{ik} = \sum_{j=1}^{n} a_{ij}b_{jk}$$

THEOREM 2.4. Associativity of Matrix Multiplication Let A, B, C be matrices such that the products AB and BC can be formed. Then $(AB)C = A(BC)$.

Proof. Let A be $m \times n$, B be $n \times p$, and C be $p \times q$. The (i, l)-entry of $(AB)C$ is:

$$\sum_{k=1}^{p} \left(\sum_{j=1}^{n} a_{ij}b_{jk} \right) c_{kl} = \sum_{j=1}^{n} a_{ij} \left(\sum_{k=1}^{p} b_{jk}c_{kl} \right)$$

which equals the (i, l)-entry of $A(BC)$. □

THEOREM 2.5. Distributivity of Matrix Multiplication Let A, B, C be matrices such that the operations are defined. Then:

$$A(B + C) = AB + AC \quad \text{and} \quad (A + B)C = AC + BC$$

Proof. Verified entrywise using the distributive law in K. □

DEFINITION 2.14. Invertible Matrix An $n \times n$ matrix A is invertible (or non-singular) if there exists an $n \times n$ matrix B such that $AB = BA = I_n$. The matrix B is called the inverse of A, denoted A^{-1}.

THEOREM 2.6. Uniqueness of Inverse If A is invertible, then its inverse is unique.

Proof. Suppose B and C are both inverses of A. Then $B = BI = B(AC) = (BA)C = IC = C$. □

THEOREM 2.7. Transpose of Product Let A, B be matrices such that AB is defined. Then ${}^t(AB) = {}^tB \cdot {}^tA$.

Proof. The (k, i)-entry of ${}^t(AB)$ is the (i, k)-entry of AB, which is $\sum_j a_{ij}b_{jk}$. The (k, i)-entry of ${}^tB \cdot {}^tA$ is $\sum_j b_{jk}a_{ij} = \sum_j a_{ij}b_{jk}$. □

3

LINEAR MAPPINGS

3.1 Mappings

DEFINITION 3.1. Mapping Let S, S' be two sets. A mapping $F : S \to S'$ is an association which to each element x of S associates an element of S', denoted by $F(x)$. The element $F(x)$ is called the image of x under F.

DEFINITION 3.2. Image of a Subset If T is a subset of S, the set of all elements $F(x)$ with $x \in T$ is called the image of T under F, denoted by $F(T)$.

DEFINITION 3.3. Composition of Mappings Let $F : S \to S'$ and $G : S' \to S''$ be mappings. The composite mapping $G \circ F : S \to S''$ is defined by $(G \circ F)(x) = G(F(x))$ for all $x \in S$.

THEOREM 3.1. Associativity of Composition Let U, V, W, S be sets and let $F : U \to V$, $G : V \to W$, $H : W \to S$ be mappings. Then:

$$H \circ (G \circ F) = (H \circ G) \circ F$$

Proof. For any $x \in U$:

$$(H \circ (G \circ F))(x) = H((G \circ F)(x)) = H(G(F(x)))$$

and:

$$((H \circ G) \circ F)(x) = (H \circ G)(F(x)) = H(G(F(x)))$$

Thus $H \circ (G \circ F) = (H \circ G) \circ F$. □

DEFINITION 3.4. Injective Mapping A mapping $f : S \to S'$ is injective (or one-to-one) if for all $x, y \in S$, $x \neq y$ implies $f(x) \neq f(y)$. Equivalently, $f(x) = f(y)$ implies $x = y$.

DEFINITION 3.5. Surjective Mapping A mapping $f : S \to S'$ is surjective (or onto) if for every $y \in S'$ there exists $x \in S$ such that $f(x) = y$.

DEFINITION 3.6. Bijective Mapping A mapping is bijective if it is both injective and surjective.

DEFINITION 3.7. Identity Mapping The identity mapping $I_S : S \to S$ is defined by $I_S(x) = x$ for all $x \in S$.

DEFINITION 3.8. Inverse Mapping Let $F : S \to S'$ be a mapping. An inverse mapping for F is a mapping $G : S' \to S$ such that:

$$G \circ F = I_S \quad \text{and} \quad F \circ G = I_{S'}$$

THEOREM 3.2. Inverse Exists iff Bijective A mapping $F : S \to S'$ has an inverse if and only if F is bijective.

Proof. ($\Rightarrow$) Suppose F has an inverse G. If $F(x) = F(y)$, then $x = G(F(x)) = G(F(y)) = y$, so F is injective. For any $z \in S'$, $z = F(G(z))$, so F is surjective.

($\Leftarrow$) Suppose F is bijective. For each $z \in S'$, there exists a unique $x \in S$ with $F(x) = z$. Define $G(z) = x$. Then G is an inverse for F. □

3.2 Linear Mappings

DEFINITION 3.9. Linear Mapping Let V, W be vector spaces over a field K. A mapping $F : V \to W$ is a linear mapping (or linear transformation) if for all $u, v \in V$ and all $c \in K$:

(LM 1) $F(u + v) = F(u) + F(v)$

(LM 2) $F(cu) = cF(u)$

THEOREM 3.3. Linear Map Determined by Basis Let V, W be vector spaces over K. Let $\{v_1, \ldots, v_n\}$ be a basis of V, and let $w_1, \ldots, w_n$ be arbitrary elements of W. Then there exists a unique linear mapping $F : V \to W$ such that $F(v_i) = w_i$ for all $i = 1, \ldots, n$.

Proof. For any $v \in V$, write $v = x_1v_1 + \cdots + x_nv_n$ uniquely. Define:

$$F(v) = x_1w_1 + \cdots + x_nw_n$$

This is well-defined by uniqueness of coordinates. We verify linearity:

For $v = \sum x_iv_i$ and $v' = \sum y_iv_i$:

$$F(v + v') = F\left(\sum(x_i + y_i)v_i\right) = \sum(x_i + y_i)w_i = \sum x_iw_i + \sum y_iw_i = F(v) + F(v')$$

For $c \in K$:

$$F(cv) = F\left(\sum cx_iv_i\right) = \sum cx_iw_i = c\sum x_iw_i = cF(v)$$

Uniqueness: If G is another linear map with $G(v_i) = w_i$, then for $v = \sum x_iv_i$:

$$G(v) = \sum x_iG(v_i) = \sum x_iw_i = F(v)$$

Thus $G = F$. □

DEFINITION 3.10. Space of Linear Maps The set of all linear mappings from V to W is denoted by $\mathcal{L}(V, W)$.

3.3 The Kernel and Image of a Linear Map

DEFINITION 3.11. Kernel Let $F : V \to W$ be a linear mapping. The kernel of F, denoted $\operatorname{Ker} F$, is the set of all $v \in V$ such that $F(v) = O$.

DEFINITION 3.12. Image The image of F, denoted $\operatorname{Im} F$, is the set of all elements of W of the form $F(v)$ for some $v \in V$.

THEOREM 3.4. Kernel is a Subspace The kernel of a linear map $F : V \to W$ is a subspace of V.

Proof. (1) $F(O) = O$, so $O \in \operatorname{Ker} F$. (2) If $u, v \in \operatorname{Ker} F$, then $F(u + v) = F(u) + F(v) = O + O = O$, so $u + v \in \operatorname{Ker} F$. (3) If $v \in \operatorname{Ker} F$ and $c \in K$, then $F(cv) = cF(v) = cO = O$, so $cv \in \operatorname{Ker} F$. □

THEOREM 3.5. Image is a Subspace The image of a linear map $F : V \to W$ is a subspace of W.

Proof. (1) $F(O) = O \in \operatorname{Im} F$. (2) If $w_1, w_2 \in \operatorname{Im} F$, there exist $v_1, v_2 \in V$ with $F(v_1) = w_1$ and $F(v_2) = w_2$. Then $w_1 + w_2 = F(v_1) + F(v_2) = F(v_1 + v_2) \in \operatorname{Im} F$. (3) If $w \in \operatorname{Im} F$ and $c \in K$, then $w = F(v)$ for some $v \in V$, so $cw = cF(v) = F(cv) \in \operatorname{Im} F$. □

THEOREM 3.6. Rank-Nullity Theorem Let V, W be vector spaces over K with V finite-dimensional, and let $F : V \to W$ be a linear map. Then:

$$\dim V = \dim \operatorname{Ker} F + \dim \operatorname{Im} F$$

Proof. Let $\{u_1, \ldots, u_m\}$ be a basis of $\operatorname{Ker} F$. Extend it to a basis $\{u_1, \ldots, u_m, v_1, \ldots, v_r\}$ of V. We claim $\{F(v_1), \ldots, F(v_r)\}$ is a basis of $\operatorname{Im} F$.

Span: For any $v \in V$, write $v = \sum a_i u_i + \sum b_j v_j$. Then:

$$F(v) = \sum a_i F(u_i) + \sum b_j F(v_j) = \sum b_j F(v_j)$$

Independence: Suppose $\sum c_j F(v_j) = O$. Then $F(\sum c_j v_j) = O$, so $\sum c_j v_j \in \operatorname{Ker} F$. Thus $\sum c_j v_j = \sum d_i u_i$ for some d_i. Rearranging: $\sum c_j v_j - \sum d_i u_i = O$. By linear independence, all $c_j = 0$ and all $d_i = 0$.

Therefore $\dim \operatorname{Im} F = r$, and $\dim V = m + r = \dim \operatorname{Ker} F + \dim \operatorname{Im} F$. □

THEOREM 3.7. Injectivity Criterion Let $F : V \to W$ be a linear map with V finite-dimensional. Then F is injective if and only if $\operatorname{Ker} F = \{O\}$.

Proof. ($\Rightarrow$) If F is injective and $v \in \operatorname{Ker} F$, then $F(v) = O = F(O)$, so $v = O$.

($\Leftarrow$) If $\operatorname{Ker} F = \{O\}$ and $F(u) = F(v)$, then $F(u - v) = O$, so $u - v \in \operatorname{Ker} F = \{O\}$. Thus $u = v$. □

3.4 Composition and Inverse of Linear Mappings

THEOREM 3.8. Composition of Linear Maps Let U, V, W be vector spaces over K. Let $F : U \to V$ and $G : V \to W$ be linear maps. Then $G \circ F : U \to W$ is linear.

Proof. For $u_1, u_2 \in U$:

$$(G \circ F)(u_1 + u_2) = G(F(u_1 + u_2)) = G(F(u_1) + F(u_2)) = G(F(u_1)) + G(F(u_2)) = (G \circ F)(u_1) + (G \circ F)(u_2)$$

For $c \in K$ and $u \in U$:

$$(G \circ F)(cu) = G(F(cu)) = G(cF(u)) = cG(F(u)) = c(G \circ F)(u)$$

Thus $G \circ F$ is linear. □

THEOREM 3.9. Inverse of Linear Map Let $F : V \to W$ be a linear map. If F has an inverse mapping $G : W \to V$, then G is linear.

Proof. Let $w_1, w_2 \in W$. Then $w_1 = F(v_1)$ and $w_2 = F(v_2)$ for unique $v_1, v_2 \in V$. We have:

$$G(w_1 + w_2) = G(F(v_1) + F(v_2)) = G(F(v_1 + v_2)) = v_1 + v_2 = G(w_1) + G(w_2)$$

Similarly, $G(cw) = cG(w)$ for $c \in K$ and $w \in W$. □

DEFINITION 3.13. Isomorphism A linear map $F : V \to W$ is an isomorphism if there exists a linear map $G : W \to V$ such that $G \circ F = I_V$ and $F \circ G = I_W$.

THEOREM 3.10. Isomorphism Criterion Let $F : V \to W$ be a linear map with V, W finite-dimensional and $\dim V = \dim W$. Then F is an isomorphism if and only if $\operatorname{Ker} F = \{O\}$ (equivalently, F is injective).

Proof. By the rank-nullity theorem, $\dim V = \dim \operatorname{Ker} F + \dim \operatorname{Im} F$. If $\operatorname{Ker} F = \{O\}$, then $\dim \operatorname{Im} F = \dim V = \dim W$, so $\operatorname{Im} F = W$. Thus F is bijective, hence an isomorphism. The converse follows from injectivity. □

3.5 Geometric Applications

DEFINITION 3.14. Line Segment Let V be a vector space and let $v_1, v_2 \in V$. The line segment between v_1 and v_2 is the set of all points $v_1 + t(v_2 - v_1)$ where $0 \leq t \leq 1$.
DEFINITION 3.15. Parallelogram Let V be a vector space and let $v_1, v_2 \in V$ be linearly independent. The parallelogram spanned by v_1, v_2 is the set of all points $t_1 v_1 + t_2 v_2$ where $0 \leq t_1 \leq 1$ and $0 \leq t_2 \leq 1$.
DEFINITION 3.16. Triangle Let V be a vector space and let $v_1, v_2 \in V$ be linearly independent. The triangle spanned by O, v_1, v_2 is the set of all points $t_1 v_1 + t_2 v_2$ where $t_1 \geq 0$, $t_2 \geq 0$, and $t_1 + t_2 \leq 1$.
DEFINITION 3.17. Convex Set A subset S of a vector space V is convex if for all $P, Q \in S$, the line segment between P and Q is contained in S.
THEOREM 3.11. Image of Convex Set Let $F : V \to W$ be a linear map and let S be a convex subset of V. Then $F(S)$ is a convex subset of W.

Proof. Let $P', Q' \in F(S)$. Then $P' = F(P)$ and $Q' = F(Q)$ for some $P, Q \in S$. For any t with $0 \leq t \leq 1$:

$$tP' + (1-t)Q' = tF(P) + (1-t)F(Q) = F(tP + (1-t)Q)$$

Since S is convex, $tP + (1-t)Q \in S$, so $F(tP + (1-t)Q) \in F(S)$. Thus $F(S)$ is convex. $\square$

4

LINEAR MAPS AND MATRICES

4.1 The Linear Map Associated with a Matrix

DEFINITION 4.1. Linear Map Associated with a Matrix Let $A = (a_{ij})$ be an $m \times n$ matrix in a field K. The linear map $L_A : K^n \to K^m$ associated with A is defined by:

$$L_A(X) = AX$$

for every column vector $X \in K^n$.

THEOREM 4.1. Matrix-Linear Map Correspondence Let A, B be $m \times n$ matrices. If $L_A = L_B$, then $A = B$.

Proof. For the standard basis vector E^j (with 1 in the j-th position and 0 elsewhere), $L_A(E^j) = AE^j$ is the j-th column of A. If $L_A = L_B$, then the j-th columns of A and B are equal for all j, so $A = B$. □

4.2 The Matrix Associated with a Linear Map

THEOREM 4.2. Matrix Representation of Linear Map Let $L : K^n \to K^m$ be a linear map. There exists a unique $m \times n$ matrix A such that $L = L_A$, i.e., $L(X) = AX$ for all $X \in K^n$.

Proof. Let $E^1, \ldots, E^n$ be the standard basis vectors of K^n. Define A to be the matrix whose j-th column is $L(E^j)$. Then for $X = x_1E^1 + \cdots + x_nE^n$:

$$L(X) = x_1L(E^1) + \cdots + x_nL(E^n) = x_1A^1 + \cdots + x_nA^n = AX$$

where A^j denotes the j-th column of A. Thus $L = L_A$. □

THEOREM 4.3. Matrix of Composition Let A be an $m \times n$ matrix and B be a $p \times m$ matrix. Then:

$$L_{BA} = L_B \circ L_A$$

Proof. For any $X \in K^n$:

$$L_{BA}(X) = (BA)X = B(AX) = L_B(AX) = L_B(L_A(X)) = (L_B \circ L_A)(X)$$

Thus $L_{BA} = L_B \circ L_A$. □

4.3 Bases, Matrices, and Linear Maps

DEFINITION 4.2. Matrix Associated with a Linear Map Relative to Bases Let V, W be vector spaces over K with $\dim V = n$ and $\dim W = m$. Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be a basis of V and $\mathcal{B}' = \{w_1, \ldots, w_m\}$ be a basis of W. Let $F : V \to W$ be a linear

map. The matrix associated with F relative to the bases $\mathcal{B}$ and $\mathcal{B}'$, denoted $M_{\mathcal{B}'}^{\mathcal{B}}(F)$, is the unique $m \times n$ matrix $A = (a_{ij})$ such that:

$$F(v_j) = \sum_{i=1}^{m} a_{ij} w_i \quad \text{for } j = 1, \ldots, n$$

THEOREM 4.4. Coordinate Transformation Let $F : V \to W$ be a linear map. Let $\mathcal{B}$ be a basis of V and $\mathcal{B}'$ be a basis of W. If $v \in V$ has coordinate vector $X_{\mathcal{B}}(v)$ relative to $\mathcal{B}$, then the coordinate vector of $F(v)$ relative to $\mathcal{B}'$ is:

$$X_{\mathcal{B}'}(F(v)) = M_{\mathcal{B}'}^{\mathcal{B}}(F) \cdot X_{\mathcal{B}}(v)$$

Proof. Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ and $\mathcal{B}' = \{w_1, \ldots, w_m\}$. Let $A = M_{\mathcal{B}'}^{\mathcal{B}}(F) = (a_{ij})$. If $v = \sum_{j=1}^{n} x_j v_j$, then:

$$F(v) = F\left(\sum_{j=1}^{n} x_j v_j\right) = \sum_{j=1}^{n} x_j F(v_j) = \sum_{j=1}^{n} x_j \sum_{i=1}^{m} a_{ij} w_i = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j\right) w_i$$

Thus the i-th coordinate of $F(v)$ relative to $\mathcal{B}'$ is $\sum_{j=1}^{n} a_{ij} x_j$, which is the i-th entry of AX. □

THEOREM 4.5. Matrix Representation of Composition Let U, V, W be vector spaces with bases $\mathcal{B}, \mathcal{B}', \mathcal{B}''$ respectively. Let $F : U \to V$ and $G : V \to W$ be linear maps. Then:

$$M_{\mathcal{B}''}^{\mathcal{B}}(G \circ F) = M_{\mathcal{B}''}^{\mathcal{B}'}(G) \cdot M_{\mathcal{B}'}^{\mathcal{B}}(F)$$

Proof. Let $A = M_{\mathcal{B}'}^{\mathcal{B}}(F)$ and $B = M_{\mathcal{B}''}^{\mathcal{B}'}(G)$. For $u \in U$ with coordinate vector X relative to $\mathcal{B}$:

$$X_{\mathcal{B}''}((G \circ F)(u)) = X_{\mathcal{B}''}(G(F(u))) = B \cdot X_{\mathcal{B}'}(F(u)) = B(AX) = (BA)X$$

Thus $M_{\mathcal{B}''}^{\mathcal{B}}(G \circ F) = BA = M_{\mathcal{B}''}^{\mathcal{B}'}(G) \cdot M_{\mathcal{B}'}^{\mathcal{B}}(F)$. □

DEFINITION 4.3. Similar Matrices Two $n \times n$ matrices M, M' over a field K are called similar if there exists an invertible matrix N such that $M' = N^{-1}MN$.

THEOREM 4.6. Change of Basis Let $F : V \to V$ be a linear map, and let $\mathcal{B}, \mathcal{B}'$ be bases of V. Then:

$$M_{\mathcal{B}'}^{\mathcal{B}'}(F) = N^{-1} M_{\mathcal{B}}^{\mathcal{B}}(F) N$$

where $N = M_{\mathcal{B}}^{\mathcal{B}'}(\mathrm{id})$ is the change of basis matrix.

Proof. Applying the composition theorem:

$$M_{\mathcal{B}'}^{\mathcal{B}'}(F) = M_{\mathcal{B}'}^{\mathcal{B}}(\mathrm{id}) M_{\mathcal{B}}^{\mathcal{B}}(F) M_{\mathcal{B}}^{\mathcal{B}'}(\mathrm{id})$$

Since $M_{\mathcal{B}'}^{\mathcal{B}}(\mathrm{id}) = (M_{\mathcal{B}}^{\mathcal{B}'}(\mathrm{id}))^{-1}$, the result follows with $N = M_{\mathcal{B}}^{\mathcal{B}'}(\mathrm{id})$. □

DEFINITION 4.4. Diagonalizable A linear map $F : V \to V$ is diagonalizable if there exists a basis $\mathcal{B}$ of V such that $M_{\mathcal{B}}^{\mathcal{B}}(F)$ is a diagonal matrix. A square matrix A is diagonalizable if the linear map L_A is diagonalizable.

THEOREM 4.7. Diagonalization Criterion Let $F : V \to V$ be a linear map and let M be its associated matrix relative to a basis $\mathcal{B}$. Then F is diagonalizable if and only if there exists an invertible matrix N such that $N^{-1}MN$ is a diagonal matrix.

Proof. If F is diagonalizable, there exists a basis $\mathcal{B}'$ such that $M_{\mathcal{B}'}^{\mathcal{B}'}(F)$ is diagonal. By the change of basis theorem, $M_{\mathcal{B}'}^{\mathcal{B}'}(F) = N^{-1}MN$ where $N = M_{\mathcal{B}}^{\mathcal{B}'}(\mathrm{id})$.

Conversely, if $N^{-1}MN = D$ is diagonal, let $\mathcal{B}'$ be the basis such that $N = M_{\mathcal{B}}^{\mathcal{B}'}(\mathrm{id})$. Then $M_{\mathcal{B}'}^{\mathcal{B}'}(F) = D$ is diagonal. □

5

SCALAR PRODUCTS AND ORTHOGONALITY

5.1 Scalar Products

DEFINITION 5.1. Scalar Product Let V be a vector space over a field K. A scalar product on V is an association which to any pair of elements v, w of V associates a scalar, denoted by $\langle v, w\rangle$, satisfying the following properties:

(SP 1) $\langle v, w\rangle = \langle w, v\rangle$ for all $v, w \in V$.

(SP 2) If u, v, w are elements of V, then $\langle u, v + w\rangle = \langle u, v\rangle + \langle u, w\rangle$.

(SP 3) If $x \in K$, then $\langle xv, w\rangle = x\langle v, w\rangle$ and $\langle v, xw\rangle = x\langle v, w\rangle$.

DEFINITION 5.2. Non-degenerate Scalar Product A scalar product is said to be non-degenerate if it also satisfies the condition: if $v \in V$ and $\langle v, w\rangle = 0$ for all $w \in V$, then $v = O$.

DEFINITION 5.3. Orthogonal Let V be a vector space with a scalar product. Elements v, w of V are orthogonal or perpendicular if $\langle v, w\rangle = 0$. We write $v \perp w$.

DEFINITION 5.4. Orthogonal Complement If S is a subset of V, we denote by $S^\perp$ the set of all elements of V which are perpendicular to all elements of S, i.e., $\langle w, v\rangle = 0$ for all $v \in S$. Then $S^\perp$ is a subspace of V called the orthogonal space of S.

DEFINITION 5.5. Positive Definite Scalar Product Let V be a vector space over $\mathbb{R}$ with a scalar product. The scalar product is positive definite if $\langle v, v\rangle \geq 0$ for all $v \in V$, and $\langle v, v\rangle > 0$ if $v \neq O$.

DEFINITION 5.6. Norm Let V be a vector space over $\mathbb{R}$ with a positive definite scalar product. The norm of an element $v \in V$ is defined by:

$$\|v\| = \sqrt{\langle v, v\rangle}$$

DEFINITION 5.7. Distance The distance between two elements v, w of V is defined to be:

$$d(v, w) = \|v - w\|$$

THEOREM 5.1. Pythagorean Theorem If v, w are perpendicular, then:

$$\|v + w\|^2 = \|v\|^2 + \|w\|^2$$

Proof. We have:

$$\|v + w\|^2 = \langle v + w, v + w\rangle = \langle v, v\rangle + 2\langle v, w\rangle + \langle w, w\rangle = \|v\|^2 + \|w\|^2$$

because $v \perp w$ implies $\langle v, w\rangle = 0$. □

THEOREM 5.2. Parallelogram Law For any v, w we have:

$$\|v+w\|^2 + \|v-w\|^2 = 2\|v\|^2 + 2\|w\|^2$$

Proof. Expanding both norms:

$$\|v+w\|^2 = \|v\|^2 + 2\langle v, w\rangle + \|w\|^2$$
$$\|v-w\|^2 = \|v\|^2 - 2\langle v, w\rangle + \|w\|^2$$

Adding gives the result. □

THEOREM 5.3. Schwarz Inequality For all $v, w \in V$ we have:

$$|\langle v, w\rangle| \leq \|v\|\|w\|$$

Proof. If $v = O$ or $w = O$, the inequality is obvious. Assume $v \neq O$ and $w \neq O$. Let $c = \frac{\langle v,w\rangle}{\langle w,w\rangle}$ so that $v - cw$ is perpendicular to w. By Pythagoras:

$$\|v\|^2 = \|v - cw\|^2 + \|cw\|^2 \geq \|cw\|^2 = c^2\|w\|^2 = \frac{\langle v, w\rangle^2}{\|w\|^2}$$

Thus $\langle v, w\rangle^2 \leq \|v\|^2\|w\|^2$, giving the result. □

THEOREM 5.4. Triangle Inequality For all $v, w \in V$:

$$\|v+w\| \leq \|v\| + \|w\|$$

Proof. We have:

$$\|v+w\|^2 = \langle v+w, v+w\rangle = \|v\|^2 + 2\langle v, w\rangle + \|w\|^2$$

By Schwarz, $\langle v, w\rangle \leq \|v\|\|w\|$, so:

$$\|v+w\|^2 \leq \|v\|^2 + 2\|v\|\|w\| + \|w\|^2 = (\|v\| + \|w\|)^2$$

Taking square roots gives the result. □

5.2 Orthogonal Bases, Positive Definite Case

DEFINITION 5.8. Orthogonal Basis Let V be a vector space with a scalar product. A basis $\{v_1, \ldots, v_n\}$ of V is an orthogonal basis if $\langle v_i, v_j\rangle = 0$ for all $i \neq j$.

DEFINITION 5.9. Orthonormal Basis An orthogonal basis is orthonormal if in addition $\langle v_i, v_i\rangle = 1$ for all i.

THEOREM 5.5. Existence of Orthogonal Basis Let V be a finite-dimensional vector space over $\mathbb{R}$, with a positive definite scalar product. Then V has an orthogonal basis.

Proof. We prove by induction on $\dim V$. If $\dim V = 1$, any nonzero vector forms an orthogonal basis. Assume the result for spaces of dimension $< n$. Let $\dim V = n$ and let $v_1 \neq O$ be in V. Let $W = v_1^\perp$ be the orthogonal complement of the space generated by v_1. Then $V = \operatorname{span}(v_1) \oplus W$. By induction, W has an orthogonal basis $\{v_2, \ldots, v_n\}$. Then $\{v_1, v_2, \ldots, v_n\}$ is an orthogonal basis of V. □

THEOREM 5.6. Dimension of Orthogonal Complement Let V be a vector space over $\mathbb{R}$ with a positive definite scalar product. Let W be a subspace. Then:

$$\dim W + \dim W^\perp = \dim V$$

and $V = W \oplus W^\perp$.

Proof. Let $\{w_1, \ldots, w_r\}$ be an orthogonal basis of W. Extend it to an orthogonal basis $\{w_1, \ldots, w_r, u_1, \ldots, u_s\}$ of V. Then $\{u_1, \ldots, u_s\}$ is a basis of $W^\perp$. Thus $\dim W^\perp = s$ and $\dim V = r + s = \dim W + \dim W^\perp$. Since $W \cap W^\perp = \{O\}$, we have $V = W \oplus W^\perp$. □

5.3 Application to Linear Equations; The Rank

DEFINITION 5.10. Row Rank and Column Rank Let A be an $m \times n$ matrix. The row rank of A is the dimension of the subspace of K^n generated by the row vectors of A. The column rank is the dimension of the subspace of K^m generated by the column vectors.

THEOREM 5.7. Row Rank Equals Column Rank Let A be an $m \times n$ matrix. Then the row rank of A equals the column rank of A. This common value is called the rank of A.

Proof. Consider the linear map $L_A : K^n \to K^m$ given by $L_A(X) = AX$. The image of L_A is the space generated by the column vectors of A, so $\dim \operatorname{Im} L_A$ equals the column rank. The kernel of L_A is the solution space of $AX = O$. By the rank-nullity theorem, $n = \dim \operatorname{Ker} L_A + \dim \operatorname{Im} L_A$. The solution space of $AX = O$ is the orthogonal complement (in K^n with the dot product) of the row space. Thus $\dim \operatorname{Ker} L_A = n - (\text{row rank})$. Therefore $n - (\text{row rank}) + (\text{column rank}) = n$, giving row rank = column rank. □

THEOREM 5.8. Rank and Dimension of Solution Space Let A be an $m \times n$ matrix. Then:

$$\operatorname{rank}(A) + \dim(\text{solution space of } AX = O) = n$$

Proof. The solution space of $AX = O$ is $\operatorname{Ker} L_A$. By the rank-nullity theorem and the fact that $\dim \operatorname{Im} L_A = \operatorname{rank}(A)$, the result follows. □

5.4 Bilinear Maps and Matrices

DEFINITION 5.11. Bilinear Map Let U, V, W be vector spaces over K. A map $g : U \times V \to W$ is bilinear if for each fixed $u \in U$, the map $v \mapsto g(u, v)$ is linear, and for each fixed $v \in V$, the map $u \mapsto g(u, v)$ is linear.

THEOREM 5.9. Matrix Representation of Bilinear Form Let $g : K^n \times K^n \to K$ be a bilinear form. Then there exists a unique $n \times n$ matrix A such that:

$$g(X, Y) = {}^tXAY$$

for all column vectors $X, Y \in K^n$.

Proof. Let $E^1, \ldots, E^n$ be the standard basis vectors. Define $a_{ij} = g(E^i, E^j)$. Then for $X = \sum x_i E^i$ and $Y = \sum y_j E^j$:

$$g(X, Y) = g\left(\sum_i x_i E^i, \sum_j y_j E^j\right) = \sum_{i,j} x_i y_j g(E^i, E^j) = \sum_{i,j} x_i a_{ij} y_j = {}^tXAY$$

where $A = (a_{ij})$. Uniqueness follows by evaluating on basis vectors. □

5.5 General Orthogonal Bases

THEOREM 5.10. Existence of Orthogonal Basis (General) Let V be a finite-dimensional vector space over the field K with a non-degenerate scalar product. Assume that $V \neq \{O\}$. Then V has an orthogonal basis.

Proof. We prove by induction on $\dim V$. If $\dim V = 1$, any nonzero vector forms an orthogonal basis. For $\dim V = n > 1$, we first show there exists $v_1 \in V$ with $\langle v_1, v_1 \rangle \neq 0$. If not, then $\langle v, v \rangle = 0$ for all v, which implies $\langle v, w \rangle = 0$ for all v, w (by expanding

$\langle v+w, v+w\rangle$), contradicting non-degeneracy. With such v_1, let $W = v_1^\perp$. Then $V = \operatorname{span}(v_1) \oplus W$. By induction, W has an orthogonal basis, giving an orthogonal basis for V. □

5.6 The Dual Space and Scalar Products

DEFINITION 5.12. Dual Space Let V be a vector space over the field K. The dual space V^* is defined to be $\mathcal{L}(V, K)$, the space of linear maps from V to K (called linear functionals).

THEOREM 5.11. Isomorphism with Dual Space Let V be a finite-dimensional vector space over K with a non-degenerate scalar product. Then the map $v \mapsto L_v$ where $L_v(w) = \langle v, w\rangle$ is an isomorphism of V with V^*.

Proof. The map $v \mapsto L_v$ is linear. If $L_v = 0$, then $\langle v, w\rangle = 0$ for all w, so $v = O$ by non-degeneracy. Thus the map is injective. Since $\dim V = \dim V^*$, it is an isomorphism. □

THEOREM 5.12. Dimension of Orthogonal Complement (General) Let V be a finite-dimensional vector space with a non-degenerate scalar product. Let W be a subspace. Then:

$$\dim W + \dim W^\perp = \dim V$$

Proof. Consider the restriction map $V^* \to W^*$ given by $\varphi \mapsto \varphi|_W$. The kernel consists of functionals vanishing on W, which corresponds to $W^\perp$ under the isomorphism $V \cong V^*$. By rank-nullity, $\dim V^* = \dim W^* + \dim W^\perp$. Since $\dim V^* = \dim V$ and $\dim W^* = \dim W$, the result follows. □

5.7 Quadratic Forms

DEFINITION 5.13. Quadratic Form Let V be a finite-dimensional vector space over the field K. Let $g = \langle\cdot,\cdot\rangle$ be a scalar product on V. The quadratic form determined by g is the function $f : V \to K$ given by $f(v) = g(v, v) = \langle v, v\rangle$.

THEOREM 5.13. Polarization Identity Let g be a symmetric bilinear form and f the associated quadratic form. Then we can recover g from f by:

$$g(v, w) = \frac{1}{2}\left[f(v+w) - f(v) - f(w)\right] = \frac{1}{4}\left[f(v+w) - f(v-w)\right]$$

Proof. Expanding $f(v+w) = \langle v+w, v+w\rangle = \langle v, v\rangle + 2\langle v, w\rangle + \langle w, w\rangle = f(v) + 2g(v, w) + f(w)$. Solving for $g(v, w)$ gives the first formula. Similarly for the second. □

5.8 Sylvester's Theorem

THEOREM 5.14. Sylvester's Theorem (Law of Inertia) Let V be a finite-dimensional vector space over $\mathbb{R}$ with a scalar product. Then there exists an integer $r \geq 0$ such that if $\{v_1, \ldots, v_n\}$ is an orthogonal basis of V, then exactly r of the values $\langle v_i, v_i\rangle$ are positive, s are negative, and t are zero, where $r + s + t = n$. The numbers r, s, t do not depend on the choice of orthogonal basis.

Proof. Let $\{v_1, \ldots, v_n\}$ and $\{w_1, \ldots, w_n\}$ be orthogonal bases. Suppose ordered so that $\langle v_i, v_i\rangle > 0$ for $i \leq r$, $\langle v_i, v_i\rangle < 0$ for $r < i \leq r+s$, and $\langle v_i, v_i\rangle = 0$ for $i > r+s$. Similarly for $\{w_j\}$ with r', s', t'.

Let V^+ be the space generated by $\{v_1, \ldots, v_r\}$ and V'^+ by $\{w_1, \ldots, w_{r'}\}$. On V^+, the form is positive definite; on the space generated by $\{v_{r+1}, \ldots, v_n\}$, the form is negative semidefinite. Thus $V^+ \cap \operatorname{span}(v_{r+1}, \ldots, v_n) = \{O\}$, so $V^+ + \operatorname{span}(v_{r+1}, \ldots, v_n)$ is a direct sum. Similarly for V'^+.

If $r' > r$, then by dimension counting, $V'^+ \cap \operatorname{span}(v_{r+1}, \ldots, v_n) \neq \{O\}$. But any nonzero vector in this intersection has $\langle v, v\rangle > 0$ (being in V'^+) and $\langle v, v\rangle \leq 0$ (being in $\operatorname{span}(v_{r+1}, \ldots, v_n)$), a contradiction. Thus $r' \leq r$. By symmetry, $r \leq r'$, so $r = r'$. Similarly $s = s'$, and thus $t = t'$. □

DEFINITION 5.14. Index of Positivity and Nullity The integer r of Sylvester's theorem is called the index of positivity. The integer t (the number of zero values) is called the index of nullity. The form is non-degenerate if and only if its index of nullity is 0.

6

DETERMINANTS

6.1 Determinants of Order 2

DEFINITION 6.1. Determinant of Order 2 Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a 2×2 matrix in a field K. The determinant of A is defined to be $ad - bc$. We denote it by:

$$\det(A) = ad - bc \quad \text{or} \quad \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

THEOREM 6.1. Properties of 2x2 Determinants The determinant satisfies the following properties:

(1) As a function of the column vectors, the determinant is linear. That is, if C, C' are column vectors and $C'' = C + C'$, then:

$$\det(C'', C^2) = \det(C, C^2) + \det(C', C^2)$$

Similarly for the second column.

(2) If two columns are equal, then the determinant is 0.

(3) The determinant of the identity matrix I is 1: $\det(I) = 1$.

(4) If two columns are interchanged, the determinant changes by a sign.

(5) If one adds a scalar multiple of one column to another, the value of the determinant does not change.

(6) $\det({}^tA) = \det(A)$ for any 2×2 matrix A.

Proof. Properties (1)-(3) follow by direct computation. Property (4) follows from (1) and (2): if C^1, C^2 are columns, then:

$$0 = \det(C^1+C^2, C^1+C^2) = \det(C^1, C^1)+\det(C^1, C^2)+\det(C^2, C^1)+\det(C^2, C^2) = \det(C^1, C^2)+\det(C^2,$$

Thus $\det(C^2, C^1) = -\det(C^1, C^2)$. Property (5) follows from (1) and (2). Property (6) is verified directly. $\square$

6.2 Existence of Determinants

DEFINITION 6.2. Determinant of Order n Let $A = (a_{ij})$ be an $n \times n$ matrix. We define the determinant by expansion according to the first row:

$$\det(A) = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det(A_{1j})$$

where A_{1j} is the $(n-1) \times (n-1)$ matrix obtained by deleting the first row and j-th column of A.

THEOREM 6.2. Uniqueness of Determinants There exists a unique function $D : \mathrm{Mat}_{n\times n}(K) \to K$ satisfying the following properties:

(D1) D is linear in each column when the other columns are fixed.

(D2) If two adjacent columns are equal, then $D(A) = 0$.

(D3) $D(I) = 1$.

This function is the determinant.

Proof. The existence is proved by induction using the expansion formula. For uniqueness, any function satisfying (D1)-(D3) must satisfy all the properties of determinants, including the alternating property and the expansion formula, which determines D uniquely. □

THEOREM 6.3. Properties of Determinants The determinant satisfies:

(1) If A has a column of zeros, then $\det(A) = 0$.

(2) If two columns are interchanged, the determinant changes by a sign.

(3) If two columns are equal, then $\det(A) = 0$.

(4) $\det(AB) = \det(A)\det(B)$.

(5) $\det({}^tA) = \det(A)$.

(6) If A is invertible, then $\det(A) \neq 0$ and $\det(A^{-1}) = \det(A)^{-1}$.

Proof. Properties (1)-(3) follow from the characterizing properties. For (4), the map $B \mapsto \det(AB)/\det(A)$ satisfies (D1)-(D3), so equals $\det(B)$. Property (5) is proved by showing $\det({}^tA)$ satisfies the same characterizing properties as $\det(A)$. Property (6) follows from $\det(AA^{-1}) = \det(I) = 1$. □

6.3 Additional Properties of Determinants

THEOREM 6.4. Effect of Column Operations

(1) If B is obtained from A by multiplying one column by a scalar c, then $\det(B) = c\det(A)$.

(2) If B is obtained from A by interchanging two columns, then $\det(B) = -\det(A)$.

(3) If B is obtained from A by adding a multiple of one column to another, then $\det(B) = \det(A)$.

Proof. These follow from linearity and the alternating property. □

THEOREM 6.5. Triangular Matrices If A is an upper or lower triangular matrix, then $\det(A)$ equals the product of the diagonal entries.

Proof. By induction on n, using the expansion formula. □

6.4 Cramer's Rule

THEOREM 6.6. Cramer's Rule Let $A^1, \ldots, A^n$ be column vectors in K^n such that $\det(A^1, \ldots, A^n) \neq 0$. Let $B \in K^n$. Then the unique solution of the system:

$$x_1 A^1 + \cdots + x_n A^n = B$$

is given by:

$$x_j = \frac{\det(A^1, \ldots, B, \ldots, A^n)}{\det(A^1, \ldots, A^n)}$$

where B occurs in the j-th place instead of A^j.

Proof. Replace the j-th column of A by B to get matrix A_j. Then:

$$\det(A_j) = \det(A^1, \ldots, \sum_i x_i A^i, \ldots, A^n) = \sum_i x_i \det(A^1, \ldots, A^i, \ldots, A^n) = x_j \det(A)$$

since all terms with $i \neq j$ have two equal columns. Thus $x_j = \det(A_j)/\det(A)$. □

THEOREM 6.7. Linear Independence Criterion The column vectors $A^1, \ldots, A^n$ of an $n \times n$ matrix A are linearly independent if and only if $\det(A) \neq 0$.

Proof. If $\det(A) \neq 0$ and $\sum x_i A^i = O$, Cramer's rule gives $x_j = 0$ for all j. Conversely, if columns are linearly independent, A is invertible, so $\det(A) \neq 0$. □

6.5 Permutations

DEFINITION 6.3. Permutation A permutation of $J_n = \{1, \ldots, n\}$ is a bijection $\sigma : J_n \to J_n$.

DEFINITION 6.4. Sign of a Permutation For a permutation σ, the sign $\epsilon(\sigma)$ is defined by:

$$\epsilon(\sigma) = \frac{\det(E^{\sigma(1)}, \ldots, E^{\sigma(n)})}{\det(E^1, \ldots, E^n)} = \det(E^{\sigma(1)}, \ldots, E^{\sigma(n)})$$

where E^i are the standard basis column vectors.

THEOREM 6.8. Properties of Sign

(1) $\epsilon(\sigma) = \pm 1$.

(2) If σ, τ are permutations, then $\epsilon(\sigma \circ \tau) = \epsilon(\sigma)\epsilon(\tau)$.

(3) If σ is a transposition (interchange of two elements), then $\epsilon(\sigma) = -1$.

Proof. (1) follows since applying a permutation to the identity matrix gives a matrix with determinant ± 1. (2) follows from $\det(AB) = \det(A)\det(B)$. (3) follows since interchanging two columns changes the sign of the determinant. □

6.6 Expansion Formula and Uniqueness of Determinants

THEOREM 6.9. General Expansion Formula Let $A = (a_{ij})$ be an $n \times n$ matrix. Then:

$$\det(A) = \sum_{\sigma} \epsilon(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}$$

where the sum is taken over all permutations σ of $\{1, \ldots, n\}$.

Proof. Write each column $A^j = \sum_i a_{ij} E^i$. By multilinearity:

$$\det(A^1, \ldots, A^n) = \sum_{i_1,\ldots,i_n} a_{i_1,1} \cdots a_{i_n,n} \det(E^{i_1}, \ldots, E^{i_n})$$

The determinant on the right is zero unless $(i_1, \ldots, i_n)$ is a permutation of $(1, \ldots, n)$, in which case it equals $\epsilon(\sigma)$. □

THEOREM 6.10. Uniqueness The determinant is the unique alternating multilinear function on the columns of $n \times n$ matrices taking value 1 on the identity matrix.

Proof. The expansion formula shows any such function must equal the determinant. □

6.7 Inverse of a Matrix

THEOREM 6.11. Formula for Inverse Let A be an invertible $n \times n$ matrix. Then:

$$A^{-1} = \frac{1}{\det(A)} (b_{ij})^t$$

where $b_{ij} = (-1)^{i+j} \det(A_{ji})$ and A_{ji} is the matrix obtained by deleting the j-th row and i-th column.

Proof. Let $B = (b_{ij})$. We verify $AB = \det(A)I$. The (i, i) entry of AB is $\sum_j a_{ij} b_{ji} = \sum_j (-1)^{i+j} a_{ij} \det(A_{ij}) = \det(A)$ by expansion along row i. The (i, k) entry for $i \neq k$ is $\sum_j (-1)^{k+j} a_{ij} \det(A_{kj})$, which is the determinant of a matrix with rows i and k equal, hence 0. □

6.8 The Rank of a Matrix and Subdeterminants

THEOREM 6.12. Rank and Subdeterminants Let A be an $m \times n$ matrix. The rank of A is r if and only if there exists an $r \times r$ submatrix with nonzero determinant, and every $(r + 1) \times (r + 1)$ submatrix has determinant 0.

Proof. The rank equals the dimension of the column space, which is the maximum number of linearly independent columns. A set of k columns is linearly independent if and only if some $k \times k$ submatrix has nonzero determinant (considering the rows spanning the same space). □

7

SYMMETRIC, HERMITIAN, AND UNITARY OPERATORS

7.1 Symmetric Operators

DEFINITION 7.1. Symmetric Operator Let V be a finite-dimensional vector space over the field K. Let $A : V \to V$ be a linear map. An operator A is symmetric (with respect to a scalar product) if:

$$\langle Av, w\rangle = \langle v, Aw\rangle$$

for all $v, w \in V$.

THEOREM 7.1. Existence of Transpose Let V be a finite-dimensional vector space over K with a non-degenerate scalar product. Let $A : V \to V$ be a linear operator. Then there exists a unique linear operator $B : V \to V$ such that:

$$\langle Av, w\rangle = \langle v, Bw\rangle$$

for all $v, w \in V$.

Proof. Given $w \in V$, the map $v \mapsto \langle Av, w\rangle$ is a linear functional. Since the scalar product is non-degenerate, there exists a unique element $Bw \in V$ such that $\langle Av, w\rangle = \langle v, Bw\rangle$ for all v. The map $w \mapsto Bw$ is linear, as can be verified directly. Uniqueness follows from the non-degeneracy of the scalar product. □

DEFINITION 7.2. Transpose The operator B in the above theorem is called the transpose of A, denoted by ${}^tA$.

THEOREM 7.2. Properties of Transpose Let V be a finite-dimensional vector space over K with a non-degenerate scalar product. Then:

(1) ${}^t(A + B) = {}^tA + {}^tB$

(2) ${}^t(cA) = c \cdot {}^tA$ for $c \in K$

(3) ${}^t(AB) = {}^tB \cdot {}^tA$

(4) ${}^t({}^tA) = A$

Proof. These follow from the defining property of the transpose and the uniqueness in the existence theorem. □

7.2 Hermitian Operators

DEFINITION 7.3. Hermitian Product Let V be a vector space over $\mathbb{C}$. A hermitian product on V is an association which to any pair of elements v, w of V associates a complex number, denoted by $\langle v, w\rangle$, satisfying the following properties:

(HP 1) $\langle v, w\rangle = \overline{\langle w, v\rangle}$ for all $v, w \in V$.

(HP 2) $\langle u, v + w\rangle = \langle u, v\rangle + \langle u, w\rangle$.

(HP 3) If $c \in \mathbb{C}$, then $\langle cv, w\rangle = c\langle v, w\rangle$ and $\langle v, cw\rangle = \bar{c}\langle v, w\rangle$.

The product is positive definite if $\langle v, v\rangle > 0$ for $v \neq O$.

THEOREM 7.3. Existence of Adjoint Let V be a finite-dimensional vector space over $\mathbb{C}$ with a positive definite hermitian product. Let $A : V \to V$ be a linear operator. Then there exists a unique linear operator $A^* : V \to V$ such that:

$$\langle Av, w\rangle = \langle v, A^*w\rangle$$

for all $v, w \in V$.

Proof. Similar to the existence of transpose, using the Riesz representation theorem for hermitian products. □

DEFINITION 7.4. Adjoint The operator A^* is called the adjoint of A.

THEOREM 7.4. Properties of Adjoint Let V be a finite-dimensional vector space over $\mathbb{C}$ with a positive definite hermitian product. Then:

(1) $(A + B)^* = A^* + B^*$

(2) $(cA)^* = \bar{c}A^*$ for $c \in \mathbb{C}$

(3) $(AB)^* = B^*A^*$

(4) $A^{**} = A$

Proof. These follow from the defining property and uniqueness. For (2), note that $\langle cAv, w\rangle = c\langle Av, w\rangle = c\langle v, A^*w\rangle = \langle v, \bar{c}A^*w\rangle$. □

DEFINITION 7.5. Hermitian Operator An operator A is hermitian if $A = A^*$.

THEOREM 7.5. Characterization of Hermitian Operators Let V be a finite-dimensional vector space over $\mathbb{C}$ with a positive definite hermitian product. An operator A is hermitian if and only if $\langle Av, v\rangle$ is real for all $v \in V$.

Proof. If $A = A^*$, then $\langle Av, v\rangle = \langle v, Av\rangle = \overline{\langle Av, v\rangle}$, so $\langle Av, v\rangle$ is real. Conversely, if $\langle Av, v\rangle$ is real for all v, then using the polarization identity, one shows $\langle Av, w\rangle = \langle v, Aw\rangle$ for all v, w, so $A = A^*$. □

7.3 Unitary Operators

DEFINITION 7.6. Unitary Operator Let V be a finite-dimensional vector space over $\mathbb{R}$ (resp. $\mathbb{C}$) with a positive definite scalar (resp. hermitian) product. A linear map $A : V \to V$ is unitary (resp. real unitary, also called orthogonal) if:

$$\langle Av, Aw\rangle = \langle v, w\rangle$$

for all $v, w \in V$.

THEOREM 7.6. Characterization of Unitary Operators Let V be as above. The following conditions on an operator A are equivalent:

(1) A is unitary.

(2) A preserves norms: $\|Av\| = \|v\|$ for all $v \in V$.

(3) A preserves unit vectors: if $\|v\| = 1$ then $\|Av\| = 1$.

(4) $A^*A = I$ (equivalently, ${}^tAA = I$ in the real case, or ${}^t\bar{A}A = I$ in the complex case).

Proof. (1) $\Rightarrow$ (2): $\|Av\|^2 = \langle Av, Av\rangle = \langle v, v\rangle = \|v\|^2$.
(2) $\Rightarrow$ (3): Immediate.
(2) $\Rightarrow$ (1): Using the polarization identity in the real case $\langle v, w\rangle = \frac{1}{4}(\|v+w\|^2 - \|v-w\|^2)$ and the complex case, we obtain $\langle Av, Aw\rangle = \langle v, w\rangle$.
(1) $\Leftrightarrow$ (4): $\langle Av, Aw\rangle = \langle v, w\rangle$ for all v, w is equivalent to $\langle A^*Av, w\rangle = \langle v, w\rangle$ for all v, w, which is equivalent to $A^*A = I$. □

THEOREM 7.7. Unitary Maps and Orthonormal Bases Let V be a finite-dimensional vector space with a positive definite scalar or hermitian product. Let $\{v_1, \ldots, v_n\}$ be an orthonormal basis of V.

(a) If A is unitary, then $\{Av_1, \ldots, Av_n\}$ is an orthonormal basis.

(b) Conversely, if $\{w_1, \ldots, w_n\}$ is another orthonormal basis and $Av_i = w_i$ for all i, then A is unitary.

Proof. (a) $\langle Av_i, Av_j\rangle = \langle v_i, v_j\rangle = \delta_{ij}$, so $\{Av_i\}$ is orthonormal, hence a basis.
(b) For $v = \sum c_i v_i$, $Av = \sum c_i w_i$. Then $\|Av\|^2 = \sum |c_i|^2 = \|v\|^2$, so A preserves norms, hence is unitary. □

8

EIGENVECTORS AND EIGENVALUES

8.1 The Characteristic Polynomial

DEFINITION 8.1. Characteristic Polynomial Let A be an $n \times n$ matrix over a field K. The characteristic polynomial of A is defined to be:

$$P_A(t) = \det(tI - A)$$

where t is an indeterminate, I is the $n \times n$ identity matrix, and $tI - A$ is the matrix with entries $t\delta_{ij} - a_{ij}$.

THEOREM 8.1. Eigenvalue Criterion A number λ is an eigenvalue of A if and only if $P_A(\lambda) = 0$, i.e., λ is a root of the characteristic polynomial.

Proof. The number λ is an eigenvalue of A if and only if there exists a nonzero vector v such that $Av = \lambda v$, equivalently $(\lambda I - A)v = O$. This happens if and only if $\lambda I - A$ is not invertible, which is equivalent to $\det(\lambda I - A) = 0$. □

THEOREM 8.2. Similar Matrices Have Same Characteristic Polynomial If A and B are similar $n \times n$ matrices, then $P_A(t) = P_B(t)$.

Proof. Let $B = M^{-1}AM$ for some invertible M. Then:

$$P_B(t) = \det(tI - B) = \det(tI - M^{-1}AM) = \det(M^{-1}(tI - A)M) = \det(M^{-1})\det(tI - A)\det(M) = P_A(t)$$

□

8.2 Eigenvectors and Eigenvalues

DEFINITION 8.2. Eigenvalue and Eigenvector Let $A : V \to V$ be a linear operator on a vector space V over K. An element $\lambda \in K$ is called an eigenvalue of A if there exists a nonzero vector $v \in V$ such that $Av = \lambda v$. The vector v is called an eigenvector of A with eigenvalue λ.

DEFINITION 8.3. Eigenspace For an eigenvalue λ of A, the set of all vectors $v \in V$ such that $Av = \lambda v$ (including the zero vector) is called the eigenspace of λ, denoted V_λ.

THEOREM 8.3. Eigenspace is a Subspace The eigenspace V_λ is a subspace of V.

Proof. If $v, w \in V_\lambda$ and $c \in K$, then $A(v + w) = Av + Aw = \lambda v + \lambda w = \lambda(v + w)$ and $A(cv) = cAv = c\lambda v = \lambda(cv)$. □

8.3 Eigenvalues and Eigenvectors of Symmetric Matrices

THEOREM 8.4. Real Eigenvalues of Symmetric Matrices Let A be a real symmetric $n \times n$ matrix. Then all eigenvalues of A are real.

Proof. Let λ be an eigenvalue of A with eigenvector $Z \in \mathbb{C}^n$. Write $Z = X + iY$ where $X, Y \in \mathbb{R}^n$. Since A is real and symmetric, $\bar{Z} = X - iY$ is also an eigenvector with eigenvalue $\bar{\lambda}$. Then:

$$\langle AZ, Z\rangle = \lambda\langle Z, Z\rangle = \lambda \sum |z_i|^2$$

Also $\langle AZ, Z\rangle = \langle Z, AZ\rangle = \bar{\lambda}\langle Z, Z\rangle$. Thus $\lambda = \bar{\lambda}$, so λ is real. □

THEOREM 8.5. Orthogonality of Eigenvectors Let A be a real symmetric matrix. Eigenvectors corresponding to distinct eigenvalues are orthogonal.

Proof. Let $Av = \lambda v$ and $Aw = \mu w$ with $\lambda \neq \mu$. Then:

$$\lambda\langle v, w\rangle = \langle Av, w\rangle = \langle v, Aw\rangle = \mu\langle v, w\rangle$$

Since $\lambda \neq \mu$, we have $\langle v, w\rangle = 0$. □

8.4 Diagonalization of a Symmetric Linear Map

THEOREM 8.6. Spectral Theorem for Symmetric Operators Let V be a finite-dimensional vector space over $\mathbb{R}$ with a positive definite scalar product. Let $A : V \to V$ be a symmetric linear map. Then V has an orthonormal basis consisting of eigenvectors of A.

Proof. By induction on $\dim V$. The base case $\dim V = 1$ is trivial. Assume true for dimensions less than n. Since A is symmetric, it has a real eigenvalue λ with eigenvector v. Let $W = v^\perp$. For $w \in W$:

$$\langle Aw, v\rangle = \langle w, Av\rangle = \lambda\langle w, v\rangle = 0$$

So W is invariant under A. By induction, W has an orthonormal basis of eigenvectors. Adding $v/\|v\|$ gives an orthonormal basis for V. □

THEOREM 8.7. Diagonalization of Symmetric Matrices Let A be a real symmetric $n \times n$ matrix. Then there exists an orthogonal matrix U such that ${}^tUAU$ is diagonal.

Proof. By the spectral theorem, A has an orthonormal basis of eigenvectors $\{v_1, \ldots, v_n\}$. Let U be the matrix with columns $v_1, \ldots, v_n$. Then U is orthogonal and ${}^tUAU$ is diagonal with eigenvalues on the diagonal. □

8.5 The Hermitian Case

THEOREM 8.8. Spectral Theorem for Hermitian Operators Let V be a finite-dimensional vector space over $\mathbb{C}$ with a positive definite hermitian product. Let $A : V \to V$ be a hermitian operator. Then V has an orthonormal basis consisting of eigenvectors of A.

Proof. Similar to the real case, using that hermitian operators have real eigenvalues. □

THEOREM 8.9. Diagonalization of Hermitian Matrices Let A be a hermitian $n \times n$ matrix. Then there exists a unitary matrix U such that U^*AU is diagonal.

Proof. Analogous to the symmetric case, using an orthonormal basis of eigenvectors with respect to the hermitian product. □

8.6 Unitary Operators

THEOREM 8.10. Spectral Theorem for Unitary Operators Let V be a finite-dimensional vector space over $\mathbb{C}$ with a positive definite hermitian product. Let $U : V \to V$ be a unitary operator. Then V has an orthonormal basis consisting of eigenvectors of U.

Proof. First note that if W is a U-invariant subspace, then $W^\perp$ is also U-invariant. For $v \in W^\perp$ and $w \in W$: $\langle Uv, w\rangle = \langle v, U^* w\rangle = \langle v, U^{-1}w\rangle$. Since W is invariant, $U^{-1}w \in W$, so $\langle v, U^{-1}w\rangle = 0$. Thus $Uv \in W^\perp$. The result follows by induction, using that eigenvalues of unitary operators have absolute value 1. $\square$

9

POLYNOMIALS AND MATRICES

9.1 Polynomials

DEFINITION 9.1. Polynomial Let K be a field. A polynomial over K is a formal expression:

$$f(t) = a_n t^n + \cdots + a_0$$

where $a_i \in K$ and t is a variable. The coefficients $a_0, \ldots, a_n$ are elements of K. If $a_n \neq 0$, then n is called the degree of f, denoted $\deg f$, and a_n is called the leading coefficient.

DEFINITION 9.2. Degree Convention We agree to the convention that $\deg 0 = -\infty$, with the rules:

$$-\infty + (-\infty) = -\infty, \quad -\infty + a = -\infty \text{ for every integer } a, \quad -\infty < a$$

THEOREM 9.1. Degree of Product Let f, g be polynomials with coefficients in K. Then:

$$\deg(fg) = \deg f + \deg g$$

Proof. Let $f(t) = a_n t^n + \cdots + a_0$ with $a_n \neq 0$ and $g(t) = b_m t^m + \cdots + b_0$ with $b_m \neq 0$. Then:

$$(fg)(t) = a_n b_m t^{n+m} + \text{terms of lower degree}$$

Since $a_n b_m \neq 0$, we have $\deg(fg) = n + m = \deg f + \deg g$. If either f or g is 0, the convention about $-\infty$ makes the assertion hold. □

DEFINITION 9.3. Root Let f be a polynomial over K. A root (or zero) of f in K is an element $\alpha \in K$ such that $f(\alpha) = 0$.

THEOREM 9.2. Factorization of Polynomials Let f be a polynomial with complex coefficients, leading coefficient 1, and $\deg f = n \geq 1$. Then there exist complex numbers $\alpha_1, \ldots, \alpha_n$ such that:

$$f(t) = (t - \alpha_1) \cdots (t - \alpha_n)$$

The numbers $\alpha_1, \ldots, \alpha_n$ are uniquely determined up to a permutation.

9.2 Polynomials of Matrices and Linear Maps

DEFINITION 9.4. Polynomial of a Matrix Let A be a square matrix with coefficients in K. Let $f \in K[t]$ with $f(t) = a_n t^n + \cdots + a_0$. We define:

$$f(A) = a_n A^n + \cdots + a_1 A + a_0 I$$

where I is the identity matrix and A^k denotes the k-th power of A.

THEOREM 9.3. Polynomial Evaluation Properties Let $f, g \in K[t]$. Let A be a square matrix with coefficients in K. Then:

(1) $(f+g)(A) = f(A) + g(A)$

(2) $(fg)(A) = f(A)g(A)$

(3) If $c \in K$, then $(cf)(A) = cf(A)$

Proof. Let $f(t) = a_n t^n + \cdots + a_0$ and $g(t) = b_m t^m + \cdots + b_0$.
(1) Without loss of generality, assume $n \geq m$ and set $b_j = 0$ for $j > m$. Then:

$$(f+g)(A) = (a_n + b_n)A^n + \cdots + (a_0 + b_0)I = f(A) + g(A)$$

(2) Let $(fg)(t) = c_{n+m}t^{n+m} + \cdots + c_0$ where $c_k = \sum_{i=0}^{k} a_i b_{k-i}$. Then:

$$f(A)g(A) = \left(\sum_{i=0}^{n} a_i A^i\right)\left(\sum_{j=0}^{m} b_j A^j\right) = \sum_{i,j} a_i b_j A^{i+j} = \sum_{k=0}^{n+m} c_k A^k = (fg)(A)$$

(3) Immediate from the definition. □

THEOREM 9.4. Existence of Annihilating Polynomial Let A be an $n \times n$ matrix in a field K. Then there exists a non-zero polynomial $f \in K[t]$ such that $f(A) = O$.

Proof. The vector space of $n \times n$ matrices over K has dimension n^2. Hence the $n^2 + 1$ matrices $I, A, A^2, \ldots, A^{n^2}$ are linearly dependent. Therefore, there exist scalars $a_0, \ldots, a_{n^2}$, not all zero, such that:

$$a_{n^2}A^{n^2} + \cdots + a_1 A + a_0 I = O$$

Let $f(t) = a_{n^2}t^{n^2} + \cdots + a_0$. Then $f(A) = O$. □

DEFINITION 9.5. Polynomial of a Linear Map Let V be a vector space over K and $A : V \to V$ be a linear operator. For a polynomial $f \in K[t]$, we define $f(A)$ in the same way as for matrices, using the rules of composition of linear maps.

THEOREM 9.5. Existence of Annihilating Polynomial for Linear Maps Let V be a finite-dimensional vector space over K and $A : V \to V$ be a linear map. Then there exists a non-zero polynomial $f \in K[t]$ such that $f(A) = 0$ (the zero map).

Proof. The space $\mathcal{L}(V, V)$ is finite-dimensional. The operators $I, A, A^2, \ldots$ cannot all be linearly independent. Hence some non-trivial linear combination equals zero. □

10

TRIANGULATION OF MATRICES AND LINEAR MAPS

10.1 Existence of Triangulation

DEFINITION 10.1. Invariant Subspace Let V be a vector space and $A : V \to V$ a linear map. A subspace W of V is called A-invariant (or invariant under A) if A maps W into itself, i.e., $AW \subseteq W$.

DEFINITION 10.2. Fan A fan for A (in V) is a sequence of subspaces $\{V_1, \ldots, V_n\}$ such that V_i is contained in V_{i+1} for each $i = 1, \ldots, n-1$, such that $\dim V_i = i$, and such that each V_i is A-invariant. Furthermore, $V = V_n$.

DEFINITION 10.3. Fan Basis Let $\{V_1, \ldots, V_n\}$ be a fan for A. A fan basis is a basis $\{v_1, \ldots, v_n\}$ of V such that $\{v_1, \ldots, v_i\}$ is a basis for V_i for each i.

THEOREM 10.1. Fan Basis Gives Triangular Matrix Let $\{v_1, \ldots, v_n\}$ be a fan basis for A. Then the matrix associated with A relative to this basis is an upper triangular matrix.

Proof. Since $AV_i \subseteq V_i$ for each i, there exist numbers a_{ij} such that:

$$\begin{aligned}
Av_1 &= a_{11}v_1 \\
Av_2 &= a_{12}v_1 + a_{22}v_2 \\
&\vdots \\
Av_i &= a_{1i}v_1 + a_{2i}v_2 + \cdots + a_{ii}v_i \\
&\vdots \\
Av_n &= a_{1n}v_1 + a_{2n}v_2 + \cdots + a_{nn}v_n
\end{aligned}$$

This means the matrix of A with respect to this basis is upper triangular with entries a_{ij}. □

DEFINITION 10.4. Triangulable A linear map $A : V \to V$ is triangulable over K if there exists a basis of V for which the associated matrix of A is triangular. An $n \times n$ matrix over K is triangulable if it is triangulable as a linear map of K^n into itself.

THEOREM 10.2. Existence of Fan (Complex Case) Let V be a finite-dimensional vector space over the complex numbers, with $\dim V \geq 1$. Let $A : V \to V$ be a linear map. Then there exists a fan of A in V.

Proof. By induction on $\dim V$. If $\dim V = 1$, the result is trivial. Assume true for dimension $n-1$. For $\dim V = n$, by the fundamental theorem of algebra, A has a nonzero eigenvector v_1. Let $V_1 = \text{span}(v_1)$. Write $V = V_1 \oplus W$. Let P_1 be the projection on V_1 and P_2 the projection on W. Then P_2A is a linear map of V into V which maps W into W. By induction, there exists a fan $\{W_1, \ldots, W_{n-1}\}$ of P_2A in W. Let $V_i = V_1 + W_{i-1}$ for $i = 2, \ldots, n$. Then $\{V_1, \ldots, V_n\}$ is a fan for A. □

COROLLARY 10.1. Triangulation Over Complex Numbers Let V be a finite-dimensional vector space over the complex numbers, with $\dim V \geq 1$. Let $A : V \to V$ be

a linear map. Then there exists a basis of V such that the matrix of A with respect to this basis is a triangular matrix.

COROLLARY 10.2. Similarity to Triangular Matrix Let M be a matrix of complex numbers. There exists a non-singular matrix B such that $B^{-1}MB$ is a triangular matrix.

DEFINITION 10.5. Markov Matrix Let $A = (a_{ij})$ be an $n \times n$ complex matrix. If the sum of the elements of each column is 1, i.e., $\sum_i a_{ij} = 1$ for each j, then A is called a Markov matrix.

THEOREM 10.3. Eigenvalue Bound for Markov Matrices Let A be a Markov matrix such that $|a_{ij}| \leq 1$ for all i, j. Then every eigenvalue of A has absolute value ≤ 1.

10.2 Theorem of Hamilton-Cayley

THEOREM 10.4. Hamilton-Cayley Theorem Let V be a finite-dimensional vector space over the complex numbers, of dimension ≥ 1, and let $A : V \to V$ be a linear map. Let P be its characteristic polynomial. Then $P(A) = O$.

Proof. By Theorem 1.2, we can find a fan for A, say $\{V_1, \ldots, V_n\}$. Let $\{v_1, \ldots, v_n\}$ be a fan basis. The matrix of A is upper triangular with diagonal entries $a_{11}, \ldots, a_{nn}$, and the characteristic polynomial is $P(t) = (t - a_{11}) \cdots (t - a_{nn})$.

We prove by induction that $(A - a_{11}I) \cdots (A - a_{ii}I)v = O$ for all $v \in V_i$. For $i = 1$, $(A - a_{11}I)v_1 = Av_1 - a_{11}v_1 = O$. For $i > 1$, any element of V_i can be written as $v' + cv_i$ with $v' \in V_{i-1}$. By induction hypothesis, $(A - a_{11}I) \cdots (A - a_{i-1,i-1}I)v' = O$. Also, $(A - a_{ii}I)v_i \in V_{i-1}$, so applying the product gives O. Thus $P(A)v = O$ for all $v \in V_n = V$. □

COROLLARY 10.3. Hamilton-Cayley for Complex Matrices Let A be an $n \times n$ matrix of complex numbers, and let P be its characteristic polynomial. Then $P(A) = O$.

COROLLARY 10.4. Hamilton-Cayley for Any Field Let V be a finite-dimensional vector space over the field K, and let $A : V \to V$ be a linear map. Let P be the characteristic polynomial of A. Then $P(A) = O$.

Proof. Take a basis of V and let M be the matrix representing A. Then $P_M = P_A$, and it suffices to prove $P_M(M) = O$. This follows from the complex case by viewing K as embedded in its algebraic closure. □

10.3 Diagonalization of Unitary Maps

THEOREM 10.5. Spectral Theorem for Unitary Operators (Alternative Proof) Let V be a finite-dimensional vector space over the complex numbers, with $\dim V \geq 1$. Assume a positive definite hermitian product on V. Let $A : V \to V$ be a unitary map. Then there exists an orthogonal basis of V consisting of eigenvectors of A.

Proof. First observe that if w is an eigenvector for A with eigenvalue λ, then $Aw = \lambda w$ and $\lambda \neq 0$ because A preserves length.

By Theorem 1.2, we can find a fan for A, say $\{V_1, \ldots, V_n\}$. Let $\{v_1, \ldots, v_n\}$ be a fan basis. We can use the Gram-Schmidt orthogonalization process to orthogonalize it to obtain an orthonormal fan basis $\{w_1, \ldots, w_n\}$. We prove by induction that each w_i is an eigenvector for A.

Since Aw_1 is contained in V_1, there exists λ_1 such that $Aw_1 = \lambda_1 w_1$, so w_1 is an eigenvector. Assume $w_1, \ldots, w_{i-1}$ are eigenvectors with nonzero eigenvalues. There exist

scalars $c_1, \ldots, c_i$ such that $Aw_i = c_1 w_1 + \cdots + c_i w_i$. Since A preserves perpendicularity, Aw_i is perpendicular to Aw_k for every $k < i$. But $Aw_k = \lambda_k w_k$, hence Aw_i is perpendicular to w_k. Thus $c_k = 0$ for $k < i$. Hence $Aw_i = c_i w_i$ with $c_i \neq 0$ because A preserves length. $\square$

COROLLARY 10.5. Unitary Diagonalization Let A be a complex unitary matrix. Then there exists a unitary matrix U such that $U^{-1}AU$ is a diagonal matrix.

11

POLYNOMIALS AND PRIMARY DECOMPOSITION

11.1 The Euclidean Algorithm

THEOREM 11.1. Euclidean Division Let f, g be polynomials over the field K, i.e., polynomials in $K[t]$, and assume $\deg g \geq 0$. Then there exist polynomials q, r in $K[t]$ such that:

$$f(t) = q(t)g(t) + r(t)$$

and $\deg r < \deg g$. The polynomials q, r are uniquely determined by these conditions.

Proof. Let $m = \deg g$. Write $f(t) = a_n t^n + \cdots + a_0$ and $g(t) = b_m t^m + \cdots + b_0$ with $b_m \neq 0$. If $n < m$, let $q = 0$, $r = f$. If $n \geq m$, let:

$$f_1(t) = f(t) - a_n b_m^{-1} t^{n-m} g(t)$$

Then $\deg f_1 < \deg f$. By induction on n, we can find polynomials q_1, r such that $f_1 = q_1 g + r$ with $\deg r < \deg g$. Then:

$$f = (a_n b_m^{-1} t^{n-m} + q_1)g + r$$

giving the desired form. For uniqueness, if $f = q_1 g + r_1 = q_2 g + r_2$, then $(q_1 - q_2)g = r_2 - r_1$. The left side has degree $\geq \deg g$ or is 0, while the right side has degree $< \deg g$. Hence both sides are 0, so $q_1 = q_2$ and $r_1 = r_2$. □

COROLLARY 11.1. Root Factor Let f be a non-zero polynomial in $K[t]$. Let $\alpha \in K$ be such that $f(\alpha) = 0$. Then there exists a polynomial $q(t)$ in $K[t]$ such that $f(t) = (t - \alpha)q(t)$.

Proof. By the Euclidean algorithm, write $f(t) = q(t)(t - \alpha) + r(t)$ where $\deg r < \deg(t - \alpha) = 1$. Hence r is constant. Since $0 = f(\alpha) = q(\alpha)(\alpha - \alpha) + r = r$, we have $r = 0$. □

COROLLARY 11.2. Number of Roots Let f be a polynomial of degree n in $K[t]$. There are at most n roots of f in K.

Proof. Otherwise, if $m > n$ and $\alpha_1, \ldots, \alpha_m$ are distinct roots, then $f(t) = (t - \alpha_1) \cdots (t - \alpha_m)g(t)$ for some polynomial g, whence $\deg f \geq m$, contradiction. □

11.2 Greatest Common Divisor

DEFINITION 11.1. Ideal of Polynomials A subset J of $K[t]$ is called an ideal if:

(1) The zero polynomial is in J.

(2) If f, g are in J, then $f + g$ is in J.

(3) If f is in J and g is an arbitrary polynomial, then gf is in J.

THEOREM 11.2. Principal Ideal Let J be an ideal of $K[t]$. Then there exists a polynomial g which is a generator of J, i.e., $J = \{qg : q \in K[t]\}$.

Proof. If $J = \{0\}$, take $g = 0$. Otherwise, let g be a polynomial in J of smallest degree. We assert g is a generator. Let f be any element of J. By the Euclidean algorithm, $f = qg + r$ with $\deg r < \deg g$. Then $r = f - qg \in J$. Since $\deg r < \deg g$, we must have $r = 0$. Hence $f = qg$, and g is a generator. □

DEFINITION 11.2. Greatest Common Divisor Let f_1, f_2 be non-zero polynomials in $K[t]$. By a greatest common divisor of f_1, f_2 we shall mean a polynomial g such that g divides f_1 and f_2, and furthermore, if h divides both f_1 and f_2, then h divides g.

THEOREM 11.3. GCD as Generator Let f_1, f_2 be non-zero polynomials in $K[t]$. Let g be a generator for the ideal generated by f_1, f_2. Then g is a greatest common divisor of f_1 and f_2.

Proof. Since f_1 lies in the ideal generated by f_1, f_2, there exists q_1 such that $f_1 = q_1 g$, whence g divides f_1. Similarly, g divides f_2. Let h be a polynomial dividing both f_1 and f_2. Write $f_1 = h_1 h$ and $f_2 = h_2 h$. Since g is in the ideal generated by f_1, f_2, there are polynomials g_1, g_2 such that $g = g_1 f_1 + g_2 f_2$, whence $g = (g_1 h_1 + g_2 h_2)h$. Consequently h divides g. □

11.3 Unique Factorization

DEFINITION 11.3. Irreducible Polynomial A polynomial p in $K[t]$ is called irreducible (over K) if it is of degree ≥ 1 and if, given a factorization $p = fg$ with $f, g \in K[t]$, then $\deg f = 0$ or $\deg g = 0$.

THEOREM 11.4. Unique Factorization of Polynomials Let f be a polynomial in $K[t]$ of degree ≥ 1. Then f can be expressed as a product $f = p_1^{m_1} \cdots p_s^{m_s}$ where $p_1, \ldots, p_s$ are irreducible polynomials with leading coefficient 1, uniquely determined up to a permutation.

Proof. (Existence) By induction on $\deg f$. If f is irreducible, write $f = cp$ with p monic irreducible. Otherwise, $f = gh$ with lower degree factors.

(Uniqueness) If $p_1^{m_1} \cdots p_s^{m_s} = q_1^{n_1} \cdots q_t^{n_t}$, then p_1 divides some q_j, so $p_1 = q_j$ since both are monic irreducible. Cancel and continue inductively. □

THEOREM 11.5. Factorization Over Complex Numbers Let f be a polynomial in $\mathbb{C}[t]$ of degree ≥ 1. Then f has a factorization:

$$f(t) = c(t - \alpha_1)^{m_1} \cdots (t - \alpha_r)^{m_r}$$

with $\alpha_i \in \mathbb{C}$ and $c \in \mathbb{C}$. The factors $(t - \alpha_i)$ are uniquely determined.

11.4 Application to the Decomposition of a Vector Space

DEFINITION 11.4. A-invariant Subspace Let V be a vector space over K and $A : V \to V$ an operator. A subspace W of V is called invariant under A (or A-invariant) if $AW \subseteq W$.

THEOREM 11.6. Decomposition by Coprime Polynomials Let $f = gh \in K[t]$ where f, g, h are polynomials of degree ≥ 1, and assume g, h are relatively prime. Let $A : V \to V$ be an operator such that $f(A) = O$. Let $W_1 = \ker g(A)$ and $W_2 = \ker h(A)$. Then $V = W_1 \oplus W_2$.

Proof. Since g, h are relatively prime, there exist r, s such that $rg + sh = 1$. For any $v \in V$: $v = r(A)g(A)v + s(A)h(A)v$. The first term is in W_2 because $h(A)r(A)g(A)v = r(A)f(A)v = O$. Similarly, the second term is in W_1. For directness, if $w \in W_1 \cap W_2$, then $w = r(A)g(A)w + s(A)h(A)w = O$. □

THEOREM 11.7. Primary Decomposition Theorem Let V be a vector space over K and $A : V \to V$ an operator. Let $P(t)$ be a polynomial such that $P(A) = O$. Let $P(t) = (t - \alpha_1)^{m_1} \cdots (t - \alpha_r)^{m_r}$ with distinct α_i. For each i, let $W_i = \ker(A - \alpha_i I)^{m_i}$. Then $V = W_1 \oplus \cdots \oplus W_r$.

Proof. Apply the previous theorem repeatedly. □

11.5 Schur's Lemma

THEOREM 11.8. Schur's Lemma Let V be a vector space over K, and let S be a set of operators of V. Assume that V is a simple S-space (no nontrivial invariant subspaces). Let $A : V \to V$ be a linear map such that $AB = BA$ for all $B \in S$. Then either A is invertible or $A = O$.

Proof. The kernel of A is S-invariant: if $v \in \ker A$ and $B \in S$, then $A(Bv) = B(Av) = O$, so $Bv \in \ker A$. Since V is simple, either $\ker A = V$ (so $A = O$) or $\ker A = \{O\}$ and $\operatorname{Im} A = V$ (so A is invertible). □

11.6 The Jordan Normal Form

DEFINITION 11.5. $(A-\alpha I)$-cyclic Let V be a vector space over $\mathbb{C}$ and $A : V \to V$ a linear map. Let $\alpha \in \mathbb{C}$. An element $v \in V$, $v \neq O$, is called $(A-\alpha I)$-cyclic with period r if $(A - \alpha I)^r v = O$ and r is the smallest positive integer with this property.

THEOREM 11.9. Jordan Normal Form Let V be a finite-dimensional vector space over $\mathbb{C}$, $V \neq \{O\}$. Let $A : V \to V$ be an operator. Then V can be expressed as a direct sum of A-invariant cyclic subspaces. Equivalently, there exists a Jordan basis such that the matrix of A consists of Jordan blocks J_i with eigenvalue α_i on the diagonal and 1's on the superdiagonal.

Proof. By the primary decomposition theorem, assume A has only one eigenvalue α. Let $B = A - \alpha I$, nilpotent with $B^r = O$ but $B^{r-1} \neq O$. Let $v \in V$ with $B^{r-1}v \neq O$. Then v is cyclic of period r. The subspace $W = \operatorname{span}\{v, Bv, \ldots, B^{r-1}v\}$ is B-invariant. Using $\dim BV + \dim \ker B = \dim V$, by induction decompose BV into cyclic subspaces and lift to V. □

12

CONVEX SETS

12.1 Definitions

DEFINITION 12.1. Convex Set Let S be a subset of $\mathbb{R}^m$. We say that S is convex if given points P, Q in S, the line segment joining P to Q is also contained in S.
DEFINITION 12.2. Line Segment The line segment joining P to Q is the set of all points $P + t(Q - P)$ with $0 \leq t \leq 1$, or equivalently $(1 - t)P + tQ$ with $0 \leq t \leq 1$.
THEOREM 12.1. Convexity of Linear Combinations Let $P_1, \ldots, P_n$ be points of $\mathbb{R}^m$. The set of all linear combinations $x_1P_1 + \cdots + x_nP_n$ with $0 \leq x_i \leq 1$ and $x_1 + \cdots + x_n = 1$, is a convex set.

Proof. Let $P = \sum x_iP_i$ and $Q = \sum y_iP_i$ be two such combinations. For $0 \leq t \leq 1$:

$$(1 - t)P + tQ = \sum((1 - t)x_i + ty_i)P_i$$

where $0 \leq (1-t)x_i+ty_i \leq 1$ and $\sum((1-t)x_i+ty_i) = (1-t)\sum x_i+t\sum y_i = (1-t)+t = 1$. □
THEOREM 12.2. Smallest Convex Set Let $P_1, \ldots, P_n$ be points of $\mathbb{R}^m$. Any convex set which contains $P_1, \ldots, P_n$ also contains all linear combinations $\sum x_iP_i$ with $0 \leq x_i \leq 1$ and $\sum x_i = 1$.
DEFINITION 12.3. Hyperplane and Half Spaces Let A be a non-zero vector in $\mathbb{R}^n$ and $c \in \mathbb{R}$. The hyperplane H defined by $A \cdot X = c$ is the set of all X such that $A \cdot X = c$. The set of all X with $A \cdot X > c$ is called an open half space. The set of all X with $A \cdot X \geq c$ is called a closed half space.
THEOREM 12.3. Hyperplane and Half Spaces are Convex A hyperplane is convex. Open and closed half spaces are convex.

Proof. If $A \cdot P = c$ and $A \cdot Q = c$, then $A \cdot ((1 - t)P + tQ) = (1 - t)A \cdot P + tA \cdot Q = c$. Similar arguments work for half spaces. □

12.2 Separating Hyperplanes

THEOREM 12.4. Separating Hyperplane Theorem Let S be a closed convex set in $\mathbb{R}^n$. Let P be a point of $\mathbb{R}^n$. Then either P belongs to S, or there exists a hyperplane H which contains P, and such that S is contained in one of the open half spaces determined by H.

Proof. Suppose $P \notin S$. Consider the continuous function $f(X) = \|X - P\|$ on the closed set S. This function has a minimum on S; let Q be a point of S such that $\|Q - P\| \leq \|X - P\|$ for all $X \in S$. Let $N = Q - P \neq O$.

We contend the hyperplane through P perpendicular to N separates P from S. For any $Q' \in S$ and $0 < t \leq 1$, by minimality:

$$\|Q - P\|^2 \leq \|Q + t(Q' - Q) - P\|^2 = \|(Q - P) + t(Q' - Q)\|^2$$

Expanding and letting $t \to 0$ gives $0 \leq (Q - P) \cdot (Q' - Q) = N \cdot (Q' - Q)$, so $Q' \cdot N \geq Q \cdot N = (P + N) \cdot N = P \cdot N + N \cdot N > P \cdot N$. Thus S is contained in the half space $X \cdot N > P \cdot N$. □

THEOREM 12.5. Supporting Hyperplane Let S be a convex set in $\mathbb{R}^n$, and let P be a boundary point of S. Then there exists a supporting hyperplane of S at P.

Proof. Let $\bar{S}$ be the closure of S. Then $\bar{S}$ is convex and P is a boundary point of $\bar{S}$. Apply Theorem 2.1. □

12.3 Extreme Points and Supporting Hyperplanes

DEFINITION 12.4. Extreme Point Let S be a convex set and let P be a point of S. We say that P is an extreme point of S if there do not exist points Q_1, Q_2 of S with $Q_1 \neq Q_2$ such that P can be written in the form $P = tQ_1 + (1 - t)Q_2$ with $0 < t < 1$.

THEOREM 12.6. Supporting Hyperplanes Contain Extreme Points Let S be a closed convex set which is bounded. Then every supporting hyperplane of S contains an extreme point.

Proof. Let H be a supporting hyperplane at a boundary point P_0. Let $T = H \cap S$. Then T is convex, closed, and bounded. We claim an extreme point of T is also an extreme point of S.

To find an extreme point of T: Among all points of T, there is at least one whose first coordinate is smallest (since T is compact). Let T_1 be the subset of T consisting of all points whose first coordinate equals this smallest one. Then T_1 is closed and bounded. Find a point of T_1 whose second coordinate is smallest, etc. Continuing, we obtain a point $P = (p_1, \ldots, p_n)$. If $P = tX + (1 - t)Y$ with $0 < t < 1$, then by minimality of coordinates, $X = Y = P$. Thus P is an extreme point. □

12.4 The Krein-Milman Theorem

DEFINITION 12.5. Convex Closure Let E be a set of points in $\mathbb{R}^n$. The convex closure of E is the intersection of all convex sets containing E, or equivalently the set of all convex combinations of points in E.

THEOREM 12.7. Krein-Milman Theorem Let S be a closed, bounded, convex set in $\mathbb{R}^n$. Then S is the smallest closed convex set containing the extreme points of S. In other words, S is the convex closure of its extreme points.

Proof. Let S' be the intersection of all closed convex sets containing the extreme points of S. Then $S' \subseteq S$. We must show $S \subseteq S'$.

Suppose $P \in S$ and $P \notin S'$. By the separating hyperplane theorem, there exists a hyperplane H passing through P, defined by $X \cdot N = c$, such that $X \cdot N > c$ for all $X \in S'$. Let $L(X) = X \cdot N$. Since S is closed and bounded, $L(S)$ is a closed interval $[a, b]$ containing c. Let H_a be the hyperplane $X \cdot N = a$. By the supporting hyperplane theorem, H_a contains an extreme point of S. This extreme point is in S', so $a > c$, contradicting $c \in [a, b]$. Thus $S \subseteq S'$. □

A
COMPLEX NUMBERS

DEFINITION A.1. Complex Numbers The complex numbers are a set of objects which can be added and multiplied, satisfying the following conditions:

(1) Every real number is a complex number, and if α, β are real numbers, then their sum and product as complex numbers are the same as their sum and product as real numbers.

(2) There is a complex number denoted by i such that $i^2 = -1$.

(3) Every complex number can be written uniquely in the form $a + bi$ where a, b are real numbers.

(4) The ordinary laws of arithmetic concerning addition and multiplication are satisfied.

DEFINITION A.2. Conjugate Let $\alpha = a+bi$ be a complex number. The conjugate of α, denoted $\bar{\alpha}$, is defined to be $a - bi$.

THEOREM A.1. Product with Conjugate For any complex number $\alpha = a + bi$:

$$\alpha\bar{\alpha} = a^2 + b^2$$

Proof. $(a + bi)(a - bi) = a^2 - (bi)^2 = a^2 - b^2i^2 = a^2 + b^2$. □

THEOREM A.2. Inverse of Complex Number If $\alpha = a+bi \neq 0$, then the inverse of α is:

$$\alpha^{-1} = \frac{\bar{\alpha}}{a^2 + b^2} = \frac{\bar{\alpha}}{|\alpha|^2}$$

Proof. $\alpha \cdot \frac{\bar{\alpha}}{a^2+b^2} = \frac{\alpha\bar{\alpha}}{a^2+b^2} = \frac{a^2+b^2}{a^2+b^2} = 1$. □

DEFINITION A.3. Absolute Value The absolute value (or modulus) of a complex number $\alpha = a_1 + ia_2$ is defined as:

$$|\alpha| = \sqrt{a_1^2 + a_2^2}$$

THEOREM A.3. Triangle Inequality for Complex Numbers If α, β are complex numbers, then:

$$|\alpha + \beta| \leq |\alpha| + |\beta|$$

Proof. This follows from the triangle inequality for the norm of vectors in $\mathbb{R}^2$, using the vector interpretation of complex numbers. □

THEOREM A.4. Fundamental Theorem of Algebra The complex numbers are algebraically closed. In other words, every polynomial $f \in \mathbb{C}[t]$ of degree ≥ 1 has a root in $\mathbb{C}$.

Proof. Let $f(t) = a_nt^n + a_{n-1}t^{n-1} + \cdots + a_0$ with $a_n \neq 0$. For every real $R > 0$, the function $|f|$ is continuous on the closed disc of radius R, hence has a minimum. When

$|t|$ becomes large, $|f(t)|$ also becomes large, so there exists $R_0 > 0$ such that if $|t| > R_0$ then $|f(t)| > C$ for any given $C > 0$.

Consequently, $|f|$ has an absolute minimum at some point z_0 in the closed disc of radius R_0. We prove $f(z_0) = 0$.

Express f in the form $f(t) = c_0 + c_1(t - z_0) + \cdots + c_n(t - z_0)^n$. If $f(z_0) = c_0 \neq 0$, let m be the smallest integer > 0 with $c_m \neq 0$. Then:

$$f(t) = c_0 + c_m z^m + z^{m+1} g(z)$$

where $z = t - z_0$. Choose z_1 with $z_1^m = -c_0/c_m$ and consider $z = \lambda z_1$ with $0 < \lambda < 1$. For sufficiently small λ, we get $|f(\lambda z_1)| < |c_0|$, contradicting the minimality of $|f(z_0)|$. Thus $f(z_0) = 0$. $\square$

B

IWASAWA DECOMPOSITION AND OTHERS

DEFINITION B.1. Group A group G is a set together with a mapping (called the product) from $G \times G$ to G, associating to each pair $(x, y) \in G$ an element $xy \in G$, satisfying the following axioms:

(GR 1) The product is associative: $(xy)z = x(yz)$ for all $x, y, z \in G$.

(GR 2) There is an element $e \in G$ such that $ex = xe = x$ for all $x \in G$ (called the unit element).

(GR 3) Given $x \in G$, there exists $x^{-1} \in G$ such that $xx^{-1} = x^{-1}x = e$ (called the inverse).

DEFINITION B.2. Subgroup Let G be a group and H a subset of G. Then H is a subgroup if it contains the unit element, is closed under taking products and inverses, i.e., if $x, y \in H$ then $xy \in H$ and $x^{-1} \in H$.

DEFINITION B.3. Special Linear Group Let $SL_n(\mathbb{R})$ denote the set of matrices with determinant 1. This forms a group under matrix multiplication, called the special linear group.

B.1 Iwasawa Decomposition

DEFINITION B.4. Iwasawa Subgroups Let $G = SL_n(\mathbb{R})$. We define:

U = subgroup of upper triangular matrices with 1's on the diagonal (called unipotent):

$$u(x) = \begin{pmatrix} 1 & x_{12} & \cdots & x_{1n} \\ 0 & 1 & \cdots & x_{2n} \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

A = subgroup of positive diagonal elements:

$$a = \begin{pmatrix} a_1 & & \\ & \ddots & \\ & & a_n \end{pmatrix} \text{ with } a_i > 0 \text{ for all } i$$

K = subgroup of real unitary matrices k satisfying ${}^t k = k^{-1}$.

THEOREM B.1. Iwasawa Decomposition The product map $U \times A \times K \to G$ given by $(u, a, k) \mapsto uak$ is a bijection.

Proof. Let $e_1, \ldots, e_n$ be the standard unit vectors. For $g = (g_{ij}) \in G$, apply the Gram-Schmidt orthogonalization process to the vectors $ge_1, \ldots, ge_n$. This yields an orthonormal basis, giving the decomposition. The upper triangular structure comes from the orthogonalization process, the diagonal elements give A, and the orthogonal transformation gives K. □

B.2 Polar Decompositions

DEFINITION B.5. Hermitian Positive Definite Matrices We denote by $\mathrm{SPos}_n(\mathbb{C})$ the set of all hermitian positive definite $n \times n$ matrices with determinant 1.

THEOREM B.2. Square Root in SPos Let $p \in \mathrm{SPos}_n(\mathbb{C})$. Then p has a unique square root in $\mathrm{SPos}_n(\mathbb{C})$.

THEOREM B.3. Quadratic Map Bijection Let $G = SL_n(\mathbb{C})$ and K be the complex unitary subgroup. The quadratic map $g \mapsto gg^*$ (where $g^* = {}^t\bar{g}$) induces a bijection:

$$G/K \to \mathrm{SPos}_n(\mathbb{C})$$

THEOREM B.4. KAK Decomposition The group $G = SL_n(\mathbb{C})$ has the decomposition (non-unique):

$$G = KAK$$

If $g \in G$ is written as $g = k_1 b k_2$ with $k_1, k_2 \in K$ and $b \in A$, then b is uniquely determined up to a permutation of the diagonal elements.

Proof. Given $g \in G$, there exists $k_1 \in K$ and $b \in A$ such that $gg^* = k_1 b^2 k_1^{-1}$ by the spectral theorem. By the bijection of the quadratic map, there exists $k_2 \in K$ such that $g = k_1 b k_2$. The uniqueness follows since b^2 is the diagonal matrix of eigenvalues of gg^*, which are uniquely determined up to permutation. □

THEOREM B.5. Polar Decomposition Let $G = SL_n(\mathbb{C})$ and $\mathbf{P} = \mathrm{SPos}_n(\mathbb{C})$. Then $G = \mathbf{P}K$, and the decomposition of an element $g = pk$ with $p \in \mathbf{P}$, $k \in K$ is unique.

Proof. The existence follows from the KAK decomposition. For uniqueness, suppose $g = pk$. Then $gg^* = pp^* = p^2$. The uniqueness of the square root shows that p is uniquely determined by g, whence so is k. □